

AVAC4QGIS 1.0.0: a QGIS framework for avalanche propagation and avalanche-generated impulse-wave simulation

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Methods checkpoint for source and Windows plugin version 1.0.0. The accepted numerical kernels are unchanged from release 0.6.1. Uncompleted observational and coupled-validation sections remain draft material.

Abstract

AVAC4QGIS is a QGIS framework for spatially explicit simulation of snow avalanche propagation and, optionally, the resulting impulse-wave forcing. AVAC advances a Cartesian, depth-averaged granular-flow model on a fixed DEM with Coulomb–Voellmy resistance, a curvature-dependent normal-load correction, finite terrain-tangent velocity transport, an analytical frozen-coefficient basal-source update, and static-yield arrest. Conservative shallow-layer correction-flux damping and synchronized CFL preflight complement the wet–dry finite-volume scheme. Voellmy source updates retain an exact zero even on a super-yield bed; the legacy pure-Coulomb fallback is kept separate. WAVE subsequently advances the standard GeoClaw shallow-water equations on a separately selected terrain/bathymetry surface. The offline coupling transfers positive avalanche volume flux and horizontal momentum across a polygon-derived shoreline. This paper specifies the implemented equations, numerical treatment, geospatial preprocessing, and physical scope of the two solvers, including the assumptions and current implementation limitations that affect interpretation of their outputs.

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1 Introduction

2 AVAC4QGIS model system

AVAC4QGIS organizes two depth-averaged solvers in a single geospatial workflow. AVAC represents avalanche release and propagation as a mobile depth layer and can be used independently. It has no erosion, entrainment, compaction, mass source, deposited-bed evolution, or changing topography; its final-time depth is therefore an arrested-layer field rather than a separate physical deposition model. When the optional impulse-wave workflow is selected, a completed AVAC calculation supplies a time-dependent source to WAVE, which represents wave generation, propagation, and inundation. The coupling is sequential and one-way: WAVE reads AVAC output, whereas the water does not alter the preceding avalanche calculation.

Both solvers are developments of GeoClaw, an open-source finite-volume code for depth-averaged geophysical flows with wetting and drying (George, 2008; Berger et al., 2011). This common numerical basis gives the two states the same conservative structure, but it does not make the two physical models interchangeable. AVAC adds a basal snow-avalanche rheology and a static-yield treatment; WAVE retains the standard water equations and Manning friction. Their shared state representation is instead used to define an explicit shoreline transfer operator.

The description below follows the same order as a simulation. We first define the geospatial inputs and the two available workflows, then state the AVAC and WAVE equations exactly as implemented, and finally define the transfer between them. This ordering distinguishes the governing equations from the QGIS and runtime infrastructure that makes them accessible.

2.1 System overview and workflow

In the standalone workflow, the user supplies a digital elevation model (DEM) and one or more avalanche-release polygons. The polygons are transformed to the DEM coordinate reference system; because AVAC solves Cartesian equations, that system must express horizontal distances in consistent metric units. AVAC4QGIS rasterizes the polygons against the DEM cells, creates the initial mobile-snow layer, writes an isolated run directory, and executes AVAC. The principal results are time-dependent flow depth and momentum, maximum depth and speed, and the final-time mobile-depth field. The latter is an arrested deposit only when the calculation has reached rest by its selected final time. No lake data or WAVE execution is required for this branch.

The impulse-wave workflow adds a terrain/bathymetry DEM, a polygon delimiting the water body, an absolute initial water-surface elevation, and WAVE hydraulic parameters. It requires a successfully completed AVAC run. The WAVE rectangular domain is the AVAC domain, while the polygon determines the initial lake and the shoreline across which avalanche flux is sampled. The WAVE terrain/bathymetry DEM must cover that rectangle and share its coordinate reference system, but it is a separately selected surface and need not equal the fixed AVAC bed. Consequently, the source and receiving calculations share a coordinate system and spatial coverage even when their terrain resolutions and solver cell sizes differ.

AVAC output frames remain immutable during preparation of the WAVE scenario. The plugin reconstructs the lake on the WAVE grid, samples the saved AVAC states along its shoreline, writes a conservative source time series, and then starts WAVE. This separation makes the avalanche result reusable: several lake levels, coupling factors, or hydraulic parameter sets can be evaluated without rerunning AVAC.

The two branches return to a common results workspace in QGIS (QGIS Development Team, 2026). Run metadata retain the materialized input layers, scientific parameters, solver configuration, source-run identifier, and runtime identity. Temporal raster layers, profiles, gauges, summary maps, and exports are therefore linked to the calculation that produced them rather than being handled as independent files.

Planned figure: AVAC4QGIS architecture and data flow

Show the standalone AVAC branch, the optional WAVE branch, the DEM and polygon inputs, the immutable completed-run handoff, the shoreline-source construction, managed runtimes, and the shared QGIS results workspace.

Figure 1: Working placeholder for AVAC4QGIS architecture and data flow.

2.2 AVAC governing equations and numerical method

2.2.1 Conserved variables, coordinates, and terrain representation

AVAC is solved on a Cartesian map grid (x, y) over a fixed bed elevation $B(x, y)$. Its conservative state is

$$\mathbf{q}_a = (h_a, m_{x,a}, m_{y,a})^\top = (h_a, h_a u_a, h_a v_a)^\top, \quad (1)$$

where h_a is vertical flow depth and (u_a, v_a) are the two horizontal, depth-averaged velocity components. This convention is important because many avalanche models instead use depth normal to the terrain. Whenever a terrain-tangent speed is reported, AVAC4QGIS reconstructs the vertical component $w_a = u_a B_x + v_a B_y$ and evaluates $U_{a,s} = \sqrt{u_a^2 + v_a^2 + w_a^2}$. This is a geometric reconstruction used by the basal-law projections and diagnostics. The conserved variables remain a vertical depth and horizontal map-plane momenta: AVAC neither stores a terrain-tangent momentum state nor an independent local frame. It does apply a discrete transport between estimated departure and arrival tangent planes, as described below.

The initial momenta are zero. Within the release polygons, the initial depth can be uniform, $h_a = d_0$, or corrected for elevation and slope. The optional elevation increment is

$$d_z = (B - z_{\text{ref}})G_z/100, \quad (2)$$

where G_z is the snow-depth change per 100 m. The optional slope factor is

$$f_s = \frac{\sin \theta_c - \nu \cos \theta_c}{\sin \theta - \nu \cos \theta}, \quad \theta = \tan^{-1}(\|\nabla B\|), \quad (3)$$

and is applied only where $\theta > 25^\circ$ and the absolute value of the denominator exceeds 10^{-12} ; else it is set to zero. Thus the four supported choices inside the release mask are d_0 , $d_0 + d_z$, $d_0 f_s$, and $(d_0 + d_z)f_s$. Outside that mask the mobile depth is zero. The cell-centred release raster is conservatively constructed in two stages. QGIS reports the selected raster extent as outer pixel edges. AVAC4QGIS therefore represents sample i, j by the common DEM and release-cell centres

$$x_i = x_{\min} + (i - \frac{1}{2})\Delta x, \quad y_j = y_{\min} + (j - \frac{1}{2})\Delta y, \quad (4)$$

for $i = 1, \dots, N_x$ and $j = 1, \dots, N_y$, while retaining $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$ as physical outer edges in terrain metadata. First, the fraction of each DEM cell occupied by the union of the release polygons is estimated on a regular 64×64 subcell grid. Exterior rings, holes, multipart features, and overlapping polygons are handled jointly, so the fraction remains between zero and one; this stage is a high-resolution area approximation rather than an exact polygon-cell intersection. Second, the resulting piecewise-constant raster is conservatively averaged onto each solver cell from the exact rectangular overlap with the source cells. If the solver spacing is $r\Delta x$, r must be an integer and both N_x and N_y must be divisible by r . A non-divisible selected DEM is rejected: silently cropping it would lose QGIS release cells, whereas extending its domain would add extrapolated terrain and qinit-free solver cells.

GeoClaw's toptype-3 reader shifts an ESRI x111 corner to the first terrain-sample centre and interpolates the supplied elevations as nodal data. AVAC4QGIS consequently writes a one-sample, replicated terrain halo only for that topography file. For a QGIS DEM extending from $x_{\min}$ to $x_{\max}$, the topography samples span from $x_{\min} - \Delta x/2$ to $x_{\max} + \Delta x/2$ (and analogously in y), which brackets every retained solver edge. The release coverage and binary initializer are not padded: the latter writes the original western and northern cell centres and native spacings, from which its reader reconstructs the original source-cell edges by adding and subtracting one half-cell. Thus the QGIS terrain cells, release cells, qinit cells, and solver rectangle are co-registered; the replicated halo supplies interpolation support rather than additional terrain or initialized mass. GeoClaw's maximum-value fixed grid (FGmax) instead specifies its first and last output *sample points*. AVAC4QGIS supplies those points at the finite-volume cell centres in Eq. (4), rather than at the physical outer edges. It consequently writes $N_x \times N_y$ peak samples; when the QGIS reader restores each sample's half-cell envelope, the maximum products cover their configured output rectangle without a half-cell shift or an extra row or column. The time-dependent fixed-grid products use their own edge-based GeoClaw convention and have the same raster registration when they share that domain. Raw FGmax files produced with the former edge-sample setup must be regenerated by rerunning: their samples do not contain the missing cell-centre values, so a GIS georeferencing change alone cannot repair them. A local fine DEM is instead an overlay on that base field. It cannot receive the external replicated halo, because GeoClaw gives the finer topography file priority wherever its nodal envelope overlaps the base terrain. AVAC4QGIS therefore writes the fine file on a half-spacing node grid whose first and last nodes lie exactly on the fine QGIS outer edges. Its edge and midpoint values sample the same replicated-edge bilinear surface within the fine footprint, while the base DEM remains active at every exterior point. Thus the local terrain is neither extrapolated over an undocumented half-cell fringe nor truncated at its own boundary. Prepared `init.xyz` and `init.avacbin` files written by earlier plugin versions must be regenerated: their format already defined the listed coordinates as cell centres, but the former writer violated that contract. AVAC4QGIS validates the release controls before evaluating the corrections and requires the unscaled candidate depth in every covered cell to be finite and non-negative. A release polygon that covers a non-finite DEM cell, a non-finite control, or a slope/elevation correction that produces a negative candidate is rejected with its affected cell location; it is not silently converted to a dry cell. The fractional coverage factor is applied only after this check, and both qinit writers independently reject non-finite or negative stored values. Thus dry-state handling cannot alter an invalid initial mass field after preparation. Terrain elevations and ASCII terrain coordinates are serialized with binary64 round-trip precision (17 significant decimal digits for elevations). This avoids introducing artificial second differences by rounding a planar DEM during preparation; it does not recover precision absent from the source DEM. It also does not eliminate roundoff or conditioning errors in GeoClaw's subsequent topographic cell integration, especially on fine AMR patches.

2.2.2 Mass and momentum equations

The active AVAC solver advances the Cartesian shallow-water balance laws

$$\partial_t h_a + \partial_x(h_a u_a) + \partial_y(h_a v_a) = 0, \quad (5)$$

$$\partial_t(h_a u_a) + \partial_x\left(h_a u_a^2 + \frac{1}{2} g h_a^2\right) + \partial_y(h_a u_a v_a) = -g h_a \partial_x B + S_{x,r} + S_{x,g}, \quad (6)$$

$$\partial_t(h_a v_a) + \partial_x(h_a u_a v_a) + \partial_y\left(h_a v_a^2 + \frac{1}{2} g h_a^2\right) = -g h_a \partial_y B + S_{y,r} + S_{y,g}. \quad (7)$$

Here g is gravitational acceleration, $\mathbf{S}_r = (S_{x,r}, S_{y,r})$ combines basal resistance with the flow-parallel steep-slope correction, and $\mathbf{S}_g = (S_{x,g}, S_{y,g})$ denotes the differential limit of the separate terrain-tangent transport operator below. The conservative fluxes, topographic source, and wet-dry Riemann problem are those of GeoClaw (George, 2008; Berger et al., 2011); the rheology and arrest logic described below are AVAC-specific.

Equations (5)–(7) are map-plane shallow-water equations rather than surface-fitted Savage–Hutter equations. Their pressure flux is hydrostatic and isotropic, corresponding to an earth-pressure coefficient of one; no

active–passive earth-pressure closure is introduced. Relief enters the conservative balance through ∇B . The directional Hessian projection κ_v affects the moving basal normal-load correction below, but does not enter the hydrostatic pressure flux and is distinct from terrain-tangent velocity transport. Standard Cartesian shallow-water gravity is too large on a steep surface because the surface gradient is a tangent rather than a sine and because only the horizontal acceleration is evolved. AVAC accounts for both projections in its flow-parallel source term following the bed-gradient variant of [Hergarten and Robl \(2015\)](#). This distinction is retained when AVAC is compared with models that express thickness and momentum in a terrain-following frame.

2.2.3 Basal rheology

AVAC offers Coulomb, Voellmy, and cohesive Voellmy resistance. The first two follow the two-parameter avalanche-resistance family introduced by [Voellmy \(1955\)](#) and subsequently used in operational avalanche calculation ([Salm et al., 1990](#); [Christen et al., 2010](#)). Because the conserved variables are defined on the map plane, the geometric quantities needed by this law are introduced explicitly. Let

$$U_h = \sqrt{u_a^2 + v_a^2}, \quad \tan \phi = \|\nabla B\|, \quad \tan \psi = -\frac{(u_a, v_a) \cdot \nabla B}{U_h}, \quad (8)$$

where ϕ is the local bed angle and ψ is the inclination of the velocity in its instantaneous direction. Thus the terrain-normal depth and terrain-tangent speed represented by the Cartesian state are $h_n = h_a \cos \phi$ and $U = U_h / \cos \psi$, respectively. The curvature sampled by motion in that direction is the projection of the fixed-bed Hessian,

$$\kappa_v = \frac{u_a^2 B_{xx} + 2u_a v_a B_{xy} + v_a^2 B_{yy}}{U_h^2}. \quad (9)$$

For the scalar basal source, AVAC evaluates B_x , B_y , B_{xx} , and B_{yy} with centred differences and B_{xy} with a four-point diagonal stencil. The transport operator has the domain-aware boundary treatment described below. No curvature smoothing is applied. The constitutive basal stress combines a Coulomb contribution, cohesion, and Voellmy resistance,

$$\tau_b = \mu \sigma_b + I_C C + I_V \frac{\rho_a g}{\xi} U^2, \quad (10)$$

where ρ_a is avalanche bulk density, μ the Coulomb coefficient, ξ the Voellmy turbulent coefficient, and C cohesion. The indicators are $(I_C, I_V) = (0, 0)$ for Coulomb, $(0, 1)$ for Voellmy, and $(1, 1)$ for cohesive Voellmy. The code does not evolve σ_b as an independent variable; its gravitational and centripetal contributions enter the projected Coulomb terms in Eqs. (12) and (16).

The Cartesian steep-slope correction, curvature-dependent normal load, and basal stress are applied together. Projecting the large-slope gravity correction onto the flow direction, as proposed by [Hergarten and Robl \(2015\)](#), gives

$$\mathbf{S}_r = -h_a \left(a + b U_h^2 \right) \frac{(u_a, v_a)}{U_h}, \quad U_h > 0, \quad (11)$$

with

$$\begin{aligned} a &= g \sin^2 \phi \tan \psi + \mu g \cos \phi \cos \psi + I_C \frac{C \cos \psi}{\rho_a h_a \cos \phi}, \\ b &= I_V \frac{g}{\xi h_a \cos \phi \cos \psi} + \mu \kappa_v \cos \phi \cos \psi. \end{aligned} \quad (12)$$

The first contribution to a removes the excessive flow-parallel component of Cartesian shallow-water gravity; the remaining terms are the projected Coulomb, cohesive, and turbulent resistances. Curvature changes b through $\mu \kappa_v \cos \phi \cos \psi$, the centripetal contribution to Coulomb normal load: concave curvature increases the load, whereas convex curvature decreases it ([Fischer et al., 2012](#)). The curvature-dependent normal-load correction is distinct from transport of a velocity vector between changing terrain tangent

planes. AVAC applies a separate finite rotation after the conservative flux update and dry-state cleanup, before the scalar basal source. Let $\mathbf{u}^*$ be the provisional horizontal velocity, $\mathbf{b}_a = \nabla B$ the arrival gradient, and H_B the local Hessian. A first-order material backtrace estimates the departure gradient as

$$\begin{aligned}\mathbf{b}_d &= \mathbf{b}_a - \Delta t H_B \mathbf{u}^*, \\ \mathbf{n}_k &= \frac{(-\mathbf{b}_k, 1)}{\sqrt{1 + \|\mathbf{b}_k\|^2}}, \quad k \in \{d, a\}.\end{aligned}\tag{13}$$

The provisional velocity is lifted to the departure tangent plane as $\mathbf{w}_d = (\mathbf{u}^*, \mathbf{b}_d \cdot \mathbf{u}^*)$. Writing $\mathbf{s} = (\mathbf{n}_d + \mathbf{n}_a)/\|\mathbf{n}_d + \mathbf{n}_a\|$, the implementation uses two orthogonal reflections to obtain the minimal rotation,

$$\begin{aligned}\mathbf{w}_a &= (I - 2\mathbf{s}\mathbf{s}^\top)(I - 2\mathbf{n}_d\mathbf{n}_d^\top)\mathbf{w}_d, \\ (m_{x,a}, m_{y,a})^\dagger &= h_a(w_{a,x}, w_{a,y}).\end{aligned}\tag{14}$$

Both graph normals point upward, so their bisector is defined for finite slopes. The rotation preserves the norm of the *lifted departure* velocity and changes neither depth nor mass. Its differential horizontal acceleration is

$$\mathbf{S}_g/h_a = -\frac{\nabla B}{1 + \|\nabla B\|^2} (\mathbf{u}^\top H_B \mathbf{u}).\tag{15}$$

Using finite normals avoids integrating this connection acceleration as a frozen quadratic growth law. The operation is an exact identity on an affine bed and at zero velocity; it is inactive for numerically dry cells and for the water model.

Transport derivatives use genuine neighbouring topography across internal patch boundaries, including periodic ghosts. At a physical exterior, the nearest wholly physical quadratic stencil supplies a gradient extrapolated to the cell centre. A direction with only two physical samples resolves its slope but not pure curvature; one sample implies extrusion. Mixed derivatives are retained when both directions have at least two samples. An affine-bed roundoff test avoids reconstructing momentum when all second differences are below the scale-aware tolerance specified below.

This is a split, local approximation using the post-flux velocity, not an exact reconstruction of all material paths entering a finite-volume cell. The preserved lifted-vector norm is not a proof of energy conservation for the complete spatial update. The hydrostatic pressure flux remains Cartesian and isotropic, and the steep-slope gravity correction remains flow-parallel. Thus the added transport does not constitute a fully surface-metric pressure and momentum discretization or a complete surface-fitted granular model.

The curvature-dependent Coulomb contribution to the speed equation can be written compactly as

$$-\mu \cos \phi \cos \psi (g + \kappa_v U_h^2).\tag{16}$$

On convex terrain, $\kappa_v < 0$, the quantity in parentheses vanishes at

$$U_{h,c} = \sqrt{-g/\kappa_v}.\tag{17}$$

AVAC does not allow this effective Coulomb normal load to become tensile, a limiting choice also documented among the curvature treatments compared by Wirbel et al. (2026). Below $U_{h,c}$ it integrates the contact branch; above it, the gravity and curvature parts of the Coulomb contribution are removed together. If a source step crosses the threshold, the crossing time is located with 60 bisection iterations and the remainder is integrated on the other branch. Voellmy drag, cohesion, and the large-slope gravity correction remain active. This is a no-tensile-normal-load approximation rather than a ballistic detachment model: AVAC does not introduce an airborne phase. The retention of cohesion outside the Coulomb-contact branch is an explicit constitutive choice of the present implementation.

On a plane with motion directly downslope, $\psi = \phi$. Combining Eq. (12) with GeoClaw's map-plane driving term and converting from U_h to U recovers

$$\frac{dU}{dt} = g \left(\sin \phi - \mu \cos \phi - I_C \frac{C}{\rho_a g h_n} - I_V \frac{U^2}{\xi h_n} \right),\tag{18}$$

and, for the non-cohesive Voellmy model, $U_\infty = [\xi h_n (\sin \phi - \mu \cos \phi)]^{1/2}$. This planar limit is used as a source-level verification of the implementation.

During one operator-split *scalar basal-source* step, h_a , the local bed gradient and Hessian, the altitude-selected rheological parameters, and the map-plane velocity direction are held fixed. Equations (11) and (12) then reduce to

$$\frac{dU_h}{dt} = -a - bU_h^2. \quad (19)$$

AVAC integrates this local equation analytically. For $a > 0$ and $b > 0$,

$$U_h^{n+1} = \begin{cases} \sqrt{\frac{a}{b}} \tan \left[\tan^{-1} \left(U_h^n \sqrt{\frac{b}{a}} \right) - \sqrt{ab} \Delta t \right], & \Phi > 0, \\ 0, & \Phi \leq 0, \end{cases} \quad (20)$$

where Φ denotes the phase in square brackets. The limiting cases are $U_h^{n+1} = \max(0, U_h^n - a\Delta t)$ for $b = 0$ and $U_h^{n+1} = U_h^n / (1 + bU_h^n \Delta t)$ for $a = 0$. Uphill motion can give $a < 0$ because the correction then returns part of GeoClaw’s excessive opposing gravity; the corresponding exact hyperbolic branch tends to $\sqrt{-a/b}$ rather than forcing a false stop. While the analytical speed is positive, the two momenta are scaled by U_h^{n+1}/U_h^n , preserving their map-plane direction. For Voellmy and cohesive Voellmy, a zero scalar result always sets both momenta to zero: no direction must be assigned to a zero vector. This can be a transient source-stage stop on a super-yield bed; the subsequent Riemann update still supplies pressure and bed-slope driving, while static equilibrium additionally requires the yield conditions below. Discarding the integrated impulse merely because the bed is super-yield would incorrectly remove the stiff $1/h_a$ Voellmy drag precisely when a source step reaches zero.

Pure Coulomb retains its separate compatibility rule: a zero scalar result sets momentum to zero only if Eq. (21) supports static rest; otherwise the incoming momentum is retained for that source substep. This is a legacy split-step fallback, not an exact continuation of the scalar ODE through zero. Consequently, although the constitutive turbulent term vanishes as $\xi \rightarrow \infty$, the two caller-level stopping policies need not coincide in that limit. No empirical speed floor or locally assigned reversal is introduced.

The closed-form impulse eliminates the explicit time-discretization error of the frozen scalar drag problem. The implementation also uses the analytical trigonometric, hyperbolic, linear, or rational branch required by all other sign combinations of a and b . For admissible non-negative rheological parameters, a $b < 0$ contact state can arise only from convex curvature; the contact-limited update transitions at Eq. (17) before that branch can become the returned momentum state. Exact integration of this frozen local subproblem does not make the flux–transport–source composition exact or independent of the time step: depth, velocity direction, geometry sampled by motion, and driving forces evolve between substeps. Timestep refinement therefore remains a separate numerical verification requirement.

Users may assign piecewise-constant values of μ , ξ , and C to altitude intervals; the moving rheological source selects the active interval from the bed elevation at the cell centre. The static Riemann checks instead select one set of parameters from the mean bed elevation at each interface, which is an intentional but discontinuous approximation at an altitude-zone boundary. The rheological source is enabled to a nominal depth of 10^6 m and therefore acts at every physically relevant avalanche depth. The angles in Eq. (8) are computed from the fixed bed rather than the evolving free surface. This is the *Gerris- z_b* choice of [Hergarten and Robl \(2015\)](#); it avoids a second, inconsistent numerical differentiation of $h_a + B$. The implementation includes the curvature-dependent normal-load and steep-slope corrections projected along the instantaneous map-plane velocity. Although B_{xy} enters both the directional curvature κ_v and the separate tangent transport, AVAC includes no surface-fitted transverse pressure or active–passive earth-pressure closure. The Riemann-interface arrest criterion described below also remains a map-plane approximation. Because the fixed DEM is differentiated without smoothing, the curvature normal-load source inherits sensitivity to DEM resolution and short-wavelength elevation noise.

2.2.4 Static yield, stopping, and arrested depth

Kinetic resistance alone is insufficient to retain a non-horizontal arrested layer: the pressure and bed-slope terms can remobilize a cell after its velocity has reached zero. AVAC therefore adds a set-valued static-yield treatment inspired by the arrest strategy used in D-Claw (George and Iverson, 2014). At the cell level the terrain angle is evaluated from centred DEM gradients. A cell that is exactly at rest remains at rest when

$$gh_a \tan \theta \leq \mu gh_a + I_C \frac{C}{\rho_a \cos^2 \theta}. \quad (21)$$

For a moving cell, the analytical update of Eq. (19) determines the source-stage speed. Voellmy and cohesive Voellmy honour a zero result even on a super-yield slope; this transient stop is not a claim that the static inequality holds. Only pure Coulomb retains the incoming momentum when a zero result occurs on a super-yield slope. Subsequent pressure and bed-slope driving remain part of the coupled flux update.

A second check acts in the normal Riemann solver to prevent pressure-driven creep through interfaces within an arrested deposit. For a wet–wet interface of spacing Δn , the directional threshold is

$$\Delta \eta_y = \left(\mu + I_C \frac{C}{\rho_a g \bar{h}_a \cos^2 \theta_n} \right) \Delta n, \quad \eta_a = h_a + B, \quad (22)$$

where $\bar{h}_a$ is the mean interface depth and θ_n is the bed angle estimated from the elevation difference normal to that interface. Before this directional test can suppress waves, each eligible wet cell must also satisfy the cell-centred map-plane condition

$$R_{ij} = \frac{\|\nabla_c \eta_a\|_2}{\mu_{ij} + I_C \frac{C_{ij}}{\rho_a g h_{a,ij} \cos^2 \theta_{B,ij}}} \leq 1, \quad \cos^2 \theta_{B,ij} = \frac{1}{1 + B_{x,c,ij}^2 + B_{y,c,ij}^2}. \quad (23)$$

Here ∇_c , $B_{x,c}$, and $B_{y,c}$ are centred differences over the four neighbouring cell centres, and μ_{ij} and C_{ij} are selected from the bed elevation of that cell. The ratio is refreshed in a transient cell-centred auxiliary field before the directional Riemann sweeps. It blocks static suppression when a diagonal free-surface gradient is vector super-yield even though each normal increment considered separately is sub-yield.

The flux is therefore suppressed only when the available wet–wet stencil is exactly at rest, its wet cells have $R_{ij} \leq 1$, and all three of the actual adjacent head jump $|\eta_{a,R} - \eta_{a,L}|$, a minmod-limited local increment, and the corresponding stencil-scale increment of η_a do not exceed $\Delta \eta_y$. The actual jump is necessary because limited or centred differences can cancel across a plateau edge or alternating cell pattern. On a flat, non-cohesive bed its condition follows from $(g/2)|h_R^2 - h_L^2| \leq \mu g \bar{h}_a \Delta n$, equivalently $|h_R - h_L| \leq \mu \Delta n$. The code also evaluates a local closed-form basal stopping predictor over the trial step, with coefficients $a_p = \mu g + I_C C / (\rho_a h_a)$ and $b_p = I_V g / (\xi h_a)$. It does not use a predicted future stop to suppress a currently moving interface: exact rest is required independently. This predictor is not the full curved-terrain source update or a source-stage CFL bound. Where the four-cell stencil is unavailable at a patch/stencil edge, the implementation falls back to the two interface states; an unavailable or invalid vector marker disables static suppression rather than applying a component-wise fallback. At an interface between an arrested wet cell and a dry cell, the same directional expression is evaluated with the wet-cell depth and the head increment $\max(0, \eta_{a,\text{wet}} - B_{\text{dry}})$, and the adjacent wet cell must meet Eq. (23). Thus cohesion is included at both wet–wet and wet–dry interfaces. These conditions preserve an admissible deposit without averaging its free surface. The normal interface threshold uses parameters selected from the mean adjacent bed elevation, whereas the vector eligibility check uses the local cell values. Both use interface/centred bed differences rather than the Hessian-based moving source, and hence remain map-plane approximations of static yield. The new vector gate is necessary for static arrest, but not a monolithic rotationally invariant two-dimensional complementarity solve: the normal interface tests remain direction-dependent, even though the wave-propagation update includes transverse corrections. States that fail these eligibility checks are allowed to evolve through the Riemann solver.

No empirical stopping velocity is used. Depths at or below the numerical dry tolerance have their momenta set to zero, and the unresolved wet–dry fringe is regularized as described below. At resolved depths, the

transition from motion to rest follows the analytical source impulse and the static-yield conditions, with the separate pure-Coulomb fallback and Voellmy transient stops stated above. This exact-zero construction for statically supportable states is the feature assessed by the deposit and steady-slope tests in Sect. 4.1. The input field historically named u_{cr} is retained in the configuration schema for compatibility but is not used by the present solver.

2.2.5 Finite-volume solution and adaptive refinement

The homogeneous and topographic parts of Eqs. (5)–(7) are advanced with GeoClaw’s augmented f -wave Riemann solver, which is well balanced for steady water surfaces and explicitly resolves inundation over variable topography (George, 2008). AVAC replaces the standard normal solver only to add the static-yield decisions above; it retains the three-wave, second-order, unsplit wave-propagation update and transverse corrections. A single user-selected TVD limiter is applied to the three wave families. A newly selected Coulomb model defaults to Minmod in the plugin, whereas Voellmy and cohesive Voellmy retain Superbee; saved explicit choices are preserved. The ISeeSnow protocol instead specifies Van Leer for its two Voellmy cases and Minmod for Coulomb. These benchmark choices are not substituted silently for the general plugin defaults.

At moving wet–dry margins, division of momentum by a vanishing cell-average depth can otherwise create a large, non-physical velocity while leaving the conserved momentum small. The Riemann solver itself uses the conservative ratio $(u_a, v_a) = (m_{x,a}, m_{y,a})/h_a$ whenever $h_a > h_{dry}$. AVAC does not project or relax cell-average momentum after each source step. Instead, on locally non-planar terrain it tapers the conservative high-order correction flux at an interface. For $h_\epsilon > 0$, let $h_{min} = \max[0, \min(h_L, h_R)]$, $r = h_{min}/h_\epsilon$, and let $\chi_f = 1$ when either neighbouring cell is classified as non-planar. The correction factor is

$$R_f = \begin{cases} \frac{\sqrt{2} r^2}{\sqrt{1 + r^4}}, & h_{min} < h_\epsilon \text{ and } \chi_f = 1, \\ 1, & \text{otherwise,} \end{cases} \quad (24)$$

Zero h_ϵ disables this taper. The default depths are $h_\epsilon = 0.05$ m for Coulomb and 0.10 m for both Voellmy laws; they are independent of mesh spacing and the reporting threshold h_v . The same factor multiplies all three components of each high-order correction and is retained when that correction is split transversely. The first-order normal flux, its Riemann states and its wave-speed calculation are unchanged. The usual $\Delta t/\Delta x$ and $\Delta t/\Delta y$ factors remain in the conservative update, rather than accumulating an extra momentum multiplier once per source step. The factor has the shape of the velocity desingularization of Kurganov and Petrova (2007), but its use on correction fluxes and its terrain trigger are AVAC-specific numerical choices, not a new physical basal law.

The two-dimensional terrain mask includes pure and mixed second differences. For stencil centre B_c , let D_2 be their largest absolute value, $A_B = \max(1, \max |B|)$ and $L_B = \max |B - B_c|$, with elevations in metres. A stencil is non-planar when

$$D_2 > \max[10^{-12} \max(1, L_B), 64\epsilon_{mach} A_B]. \quad (25)$$

This auxiliary mask is refreshed before the Riemann sweeps, including the extended ghost stencil used by the positivity relimiter. It trusts internal and periodic ghost terrain and excludes only exterior physical closures, using the same domain-aware stencil convention as the transport operator. An affine discrete bed below this roundoff threshold bypasses the taper; that property is not a guarantee that every AMR representation of an analytically planar DEM remains below the threshold after topographic integration. Water does not use the granular mask or correction taper.

Donor-based positivity relimiting is enabled for granular runs. Available cell depth bounds outgoing face fluxes, with the same positive scaling for all conserved components and the two-sided source-related flux differences retained. Face ownership is determined before any component is modified, and the extended normal and transverse stencils are evaluated consistently. The flux-based operation is distinct from a

post-update transfer of material between neighbouring cells: AVAC performs no such deficit redistribution. Residual negative depths are clipped to zero after the flux update, and numerically dry cells have zero momentum. Such clipping can add a residual volume, so conservation must still be measured rather than inferred from the relimiter alone. The default dry tolerance is $h_{\text{dry}} = 1 \times 10^{-4}$ m.

Peak diagnostics apply a separate, non-invasive depth rule. Depth is copied from the non-negative current state without applying the velocity threshold; reported velocity is zero for $h_a \leq h_v$, follows the desingularization

$$\hat{u}_a = \frac{\sqrt{2} h_a m_{x,a}}{[h_a^4 + \max(h_a^4, (4h_v)^4)]^{1/2}}, \quad \hat{v}_a = \frac{\sqrt{2} h_a m_{y,a}}{[h_a^4 + \max(h_a^4, (4h_v)^4)]^{1/2}} \quad (26)$$

for $h_v < h_a < 4h_v$, and is exactly momentum divided by depth for $h_a \geq 4h_v$. The default h_v is 0.05 m, so the diagnostic transition ends at 0.20 m. The horizontal diagnostic velocity is converted to the terrain-tangent magnitude using $w_a = u_a B_x + v_a B_y$ before the temporal maximum is updated. This reporting operation does not alter the conserved state. AVAC writes the shared GeoClaw speed guard as the deliberately nonphysical value 10^{99} m s^{-1} ; this is not an operational finite speed cap.

Topographic balance is incorporated in the augmented f -wave Riemann solve. After the flux update and dry-state cleanup, terrain-tangent transport is applied once, followed by the scalar granular rheological source, using first-order Godunov splitting. Second-order spatial reconstruction therefore does not imply a second-order accurate full source-split method.

Before committing an AMR-level step, a CFL preflight repeats the preparation and normal Riemann sweeps on private copies of the incoming depth, momentum and auxiliary fields. Its wave-speed reduction matches that of the physical flux update. If the maximum trial Courant number C_{trial} exceeds the configured ceiling C_{max} , the common level step is reduced toward

$$\Delta t_{\text{retry}} = \Delta t \frac{\min(C_{\text{target}}, 0.9C_{\text{max}})}{C_{\text{trial}}}, \quad (27)$$

and the trial is repeated. No rejected trial commits solution fields, flux registers, clocks, gauges or peak observations. The accepted physical update then executes once. A non-finite trial CFL, invalid timestep or CFL settings, a non-advancing time step, an excessive fixed step, or more than 20 rejected trials cause an explicit failure. This is an incoming-wave CFL test, not a full post-flux or post-source state acceptance test. It neither bounds every shallow-cell velocity nor proves stability or convergence of the split scheme.

The plugin defaults are $C_{\text{target}} = 0.5$ and $C_{\text{max}} = 1$. The ISeeSnow Coulomb protocol explicitly uses target 0.25, while the two Voellmy cases use 0.5, all with ceiling 1. A halved target need not halve every accepted step, because wave speeds and output/AMR synchronization also influence the selected time step.

Ordinary granular runs use five ghost-cell layers; water uses two. The synthetic Coulomb analytical harness can explicitly request a restricted two-layer compatibility configuration for its narrow strips. Real-terrain runs use a one-way, zero-gradient outflow boundary: an outward state is copied while attempted inward normal momentum is reflected to block external avalanche inflow. Verification problems may instead select the boundary condition required by their analytical solution. Wetting and drying follow GeoClaw's depth tolerance, supplemented by the AVAC arrested-edge criterion.

AVAC can use patch-based adaptive mesh refinement following the Clawpack wave-propagation framework (Berger and LeVeque, 1998; Berger et al., 2011). The plugin disables the usual free-surface wave-height flag and uses the following automatic velocity-based flag for a wet cell when its speed exceeds the level-specific tolerance and its kinetic proxy satisfies

$$\sqrt{h_a} U_h > \sqrt{h_v} U_{\text{tol}}. \quad (28)$$

The second condition prevents a vanishing-depth numerical tail from creating fine patches, and it is independent of the disabled velocity ceiling. Explicit AMR regions can also require refinement of static or dry terrain. The current plugin supplies 0.05 m s^{-1} as the speed tolerance for every requested level

transition, regrid periodically, and subcycles finer levels in time. When a fine-level CFL retry shortens the admissible step, the remaining parent interval is repartitioned into an integer number of equal substeps so that synchronization is retained. Counts are checked for finite values, integer overflow and representable time advancement; no arbitrary 100-substep ceiling replaces the numerical timestep constraint. The bundled plugin template defaults to one refinement level, and the ISeeSnow comparisons reported here also use one level; the analytical AMR tests use explicitly different level and stencil settings.

During coarse-to-fine transfer, AVAC reconstructs mobile depth rather than a lake-at-rest free surface because h_a is avalanche mass per unit map area. Minmod-limited depth slopes in the two coordinate directions are scaled together, when necessary, so that every child depth is non-negative while the parent mean is retained. During fine-to-coarse restriction, depth and both momenta are averaged over all child cells, including dry children; averaging only wet cells would duplicate mass at a partly wet front. A deliberately low sea-level value of -10^4 m prevents standard GeoClaw initialization of a new fine patch from creating water-like depth over below-datum dry topography. Together with release mapping and the exact frozen-coefficient friction impulse, these transfers make avalanche volume the primary AMR invariant, subject to the residual clipping possible in the post-flux positivity repair. The round-trip terrain writer and affine-bypass tests do not establish exact planar preservation under every AMR topographic integration: this remaining geometric conditioning issue is separate from the conservative depth and momentum transfer rules.

2.3 WAVE governing equations and numerical method

2.3.1 Shallow-water equations

WAVE uses the standard GeoClaw state $\mathbf{q}_w = (h_w, h_w u_w, h_w v_w)^T$, where h_w is water depth and (u_w, v_w) is the horizontal depth-averaged water velocity. Away from the AVAC source it solves

$$\partial_t h_w + \partial_x (h_w u_w) + \partial_y (h_w v_w) = 0, \quad (29)$$

$$\partial_t (h_w u_w) + \partial_x \left(h_w u_w^2 + \frac{1}{2} g h_w^2 \right) + \partial_y (h_w u_w v_w) = -g h_w \partial_x B_w + S_{x,M}, \quad (30)$$

$$\partial_t (h_w v_w) + \partial_x (h_w u_w v_w) + \partial_y \left(h_w v_w^2 + \frac{1}{2} g h_w^2 \right) = -g h_w \partial_y B_w + S_{y,M}, \quad (31)$$

where B_w is the terrain/bathymetry surface prepared for WAVE. It is read from a separately selected DEM that covers the AVAC rectangle in the same coordinate reference system and therefore need not equal the fixed AVAC bed B . Both standard configurations use $g = 9.81 \text{ m s}^{-2}$. S_M is Manning resistance. The coupling source introduced in Sect. 2.4 is added to all three equations at shoreline cells.

These nonlinear shallow-water equations assume hydrostatic pressure and omit frequency dispersion. They are appropriate when the relevant wavelength is large relative to water depth, but they cannot resolve the three-dimensional, non-hydrostatic impact zone. This limitation is especially relevant to landslide- and avalanche-generated waves (Poulain et al., 2023); the purpose of the present coupling is to prescribe depth-integrated mass and momentum to a long-wave propagation model, not to reproduce grain-scale impact physics.

2.3.2 Initial lake, friction, and wetting/drying

Let ζ_0 be the absolute initial lake level. Before shoreline stabilization, the desired lake-at-rest state is

$$h_w(x, y, 0) = \begin{cases} \max(\zeta_0 - B_w(x, y), 0), & (x, y) \in \Omega_L, \\ 0, & (x, y) \notin \Omega_L, \end{cases} \quad u_w = v_w = 0, \quad (32)$$

where Ω_L is the lake polygon rasterized on the WAVE mesh. When GDAL rasterization is available, the polygon uses an all-touched rule at the terrain resolution; the dependency-free fallback uses centre-point inclusion instead. If the WAVE grid is coarser, the fine binary mask is aggregated and a WAVE cell belongs to the lake when at least half of its fine mask cells are inside. This is an area approximation that prevents a narrow polygon sliver from turning an entire coarse cell into water.

GeoClaw cell-averages a piecewise-linear topographic representation. A dry cell beside a deep lake can therefore acquire a mean bed below ζ_0 even when its centre lies outside Ω_L , producing a numerical wave when the initial dry mask is released. AVAC4QGIS avoids that artifact by raising every exterior terrain sample below the waterline, and—when at least one wet interior cell remains—an eight-neighbour one-cell inner rim, to $\zeta_0 + h_{\text{dry}}$. The force-dry file is configured through $t_0 + 0.5$ s when initial or newly refined patches are constructed; in the standard one-level WAVE workflow it therefore acts only at initialization. The stabilized rim is not a permanent wall: subsequent inflow or runup can wet a cell once its computed water surface exceeds the guarded bed elevation.

WAVE expresses bed friction through Strickler coefficients $K = 1/n$, where n is the Manning coefficient. With $\mathbf{m}_w = (h_w u_w, h_w v_w)$, the implemented local source update is

$$\gamma_M = \frac{gn^2 \|\mathbf{m}_w\|}{h_w^{7/3}}, \quad \mathbf{m}_w^{n+1} = \frac{\mathbf{m}_w^n}{1 + \gamma_M \Delta t}. \quad (33)$$

It is applied only where h_w is above the dry tolerance and below the configured friction-depth limit. Two coefficients can be separated by the initial water elevation. Their executable mapping is

$$n(B_w) = \begin{cases} 1/K_w, & B_w < \zeta_0, \\ 1/K_l, & B_w \geq \zeta_0, \end{cases} \quad (34)$$

where K_w and K_l are the water and land Strickler coefficients, respectively. AVAC4QGIS writes them in this order because GeoClaw applies its first coefficient below the elevation break and its second coefficient above the break.

Because the coupled signal can be smaller than an imbalance created during lake preparation, the unforced lake is itself a model test. A stationary lake, stable wet–dry edge, and negligible drift in water volume are required before an avalanche source is interpreted; these properties are evaluated in Sect. 4.2.

2.3.3 Numerical solution and adaptive refinement

WAVE uses GeoClaw’s unmodified augmented normal Riemann solver and transverse solver (George, 2008). As in AVAC, the finite-volume update is second-order, unsplit, and based on three limited f -waves; topographic balance is embedded in the augmented f -wave solve, while the Manning and injected shoreline sources are applied by Godunov splitting. The time step is adapted to user-specified target and maximum Courant numbers. The standard configuration uses zero-order extrapolation at the four outer boundaries, while analytical tests can select periodic or wall boundaries when required. Dry states, shoreline motion, and inundation use the standard GeoClaw wetting-and-drying algorithm (Berger et al., 2011).

The underlying solver and the internal shoreline source are compatible with patch-based AMR because total source rates are divided by the area of the receiving patch cell. However, the normal AVAC4QGIS WAVE workflow in the version documented here deliberately prepares one refinement level. Thus the reported plugin simulations use a uniform WAVE grid, whereas AMR capability remains a property to be activated and evaluated separately. Verification is still required for the plugin-specific lake mask, shoreline stabilization, and source routine even though the homogeneous water solver is established GeoClaw software.

2.4 One-way AVAC-to-WAVE shoreline coupling

2.4.1 Lake-mask and shoreline construction

The coupling follows the long-standing idea of representing avalanche entry as a forcing of a depth-averaged reservoir model (Vila, 1987; Ancy, 2023). AVAC4QGIS makes that forcing spatially and temporally explicit on the WAVE mesh. The lake polygon defines the initially wet body Ω_L , not the rectangular calculation domain. Every cell in the polygon-derived WAVE mask adjacent to a cell outside that mask contributes one grid-aligned shoreline face. Such a target can be a zero-depth cell of the numerical stabilization rim

rather than an initially wet cell. For each face f , the preprocessing stores the face centre, the centre of the source-target cell, the face length L_f , and the unit normal $\mathbf{n}_f = (n_{x,f}, n_{y,f})$ pointing from land into the lake.

This construction produces a staircase approximation of the polygon at the chosen WAVE resolution. A mask corner cell can therefore contribute two faces, and all of their fluxes are retained. The shoreline resolution is controlled by WAVE cell size rather than by an independent number of sampling points, which keeps the geometric flux integral and its receiving cells in the same discretization. The polygon should retain at least one dry WAVE-cell margin to the rectangular domain boundary: a missing exterior neighbour is currently treated as dry and would otherwise create an outer-boundary source face.

2.4.2 Sampling the AVAC state

At every saved AVAC time, the three conserved components are interpolated at the WAVE shoreline-face centres. Bilinear interpolation is performed within each AVAC patch, with finite values on finer patches replacing coarser ones. Samples outside every AVAC patch are recorded and replaced by a zero state; the workflow fails only if every shoreline sample is outside AVAC coverage in every saved frame. Denoting avalanche depth and map-grid velocity by h_a and (u_a, v_a) , respectively, the inward normal volume flux per unit shoreline length is

$$q_{n,f} = (h_a u_a)_f n_{x,f} + (h_a v_a)_f n_{y,f}. \quad (35)$$

Only finite samples with $h_a > \epsilon_h$ and $q_{n,f} > \epsilon_q$ contribute to the WAVE source. The current implementation assigns the same numerical tolerance to ϵ_h and ϵ_q in their respective units. The positive-flux condition admits avalanche material entering the lake but does not withdraw water when the sampled avalanche velocity points out of it.

2.4.3 Transferred volume and horizontal momentum

For a shoreline face of length L_f , AVAC4QGIS constructs the water-volume rate

$$Q_{w,f} = \beta q_{n,f} L_f, \quad (36)$$

and supplies WAVE with the three conservative rates

$$\mathbf{R}_f = (Q_{w,f}, Q_{w,f} u_{a,f}, Q_{w,f} v_{a,f})^T. \quad (37)$$

The units of the three entries are $\text{m}^3 \text{s}^{-1}$, $\text{m}^4 \text{s}^{-2}$, and $\text{m}^4 \text{s}^{-2}$, respectively. Multiplication of the last two time integrals by water density converts them into physical horizontal momentum.

The coefficient β combines the snow-to-water volume conversion and any unresolved transfer efficiency. It can be written as

$$\beta = \varepsilon_T \frac{\rho_a}{\rho_w}, \quad (38)$$

where ρ_w is water density and ε_T is a dimensionless efficiency. For $\varepsilon_T = 1$, the avalanche mass rate $\rho_a q_{n,f} L_f$ equals the injected water mass rate $\rho_w Q_{w,f}$, and the two horizontal momentum rates also agree. For $\varepsilon_T < 1$, both transferred mass and momentum are reduced by the same factor. The current interface exposes their product β as a single “snow-to-water damping” parameter; mass and momentum conservation are therefore exact only when that parameter is set to ρ_a/ρ_w . The QGIS control supplies β directly under the name “damping” and constrains it to $0 \leq \beta \leq 0.4$.

This density dependence is not cosmetic. Laboratory work on snow-avalanche impulse waves shows that buoyancy, impact velocity, and the density ratio control momentum and energy transfer (Zitti et al., 2016). The present operator is a deliberately reduced closure: it preserves the incoming depth-integrated horizontal quantities when Eq. (38) is conservative, but it does not resolve the underlying two-phase interaction. In particular, the operator transfers no energy, vertical momentum, granular-volume removal, or pressure impulse.

2.4.4 Temporal interpolation and injection into WAVE

Equation (37) is evaluated at every saved AVAC output time. Contributions from faces that point to the same WAVE source-target cell are summed, producing a piecewise-linear time series $\mathbf{R}_c(t)$ for each source cell c . During an accepted WAVE step from t to $t + \Delta t$, the source routine samples that series at the midpoint and applies

$$\mathbf{q}_{w,c}^* = \mathbf{q}_{w,c}^n + \frac{\Delta t}{A_c} \mathbf{R}_c\left(t + \frac{1}{2}\Delta t\right), \quad (39)$$

where A_c is the area of the WAVE patch cell that receives the source. The Manning update follows within the same split source step. Because the stored quantities are total rates and division uses the actual receiving area, the integrated injection is independent of cell area and remains well defined if WAVE AMR is enabled later.

The midpoint rule is exact for a WAVE step lying within one linearly interpolated AVAC-frame interval. The implementation does not split a WAVE step at a frame-time knot, so a step spanning a change in the interpolated slope has an additional quadrature error. The AVAC output interval and its alignment with WAVE time steps are therefore coupling-resolution parameters, not merely visualization choices. The coupled verification in Sect. 4.3 examines both temporal sampling and spatial shoreline resolution.

2.4.5 Transfer ledger and provenance

Each prepared WAVE scenario retains the source AVAC run and frame times, all shoreline faces, the faces that are active in at least one frame, the receiving source cells, and the count of samples replaced because they were outside AVAC coverage. In addition, AVAC4QGIS integrates the sum of all stored source rates by the trapezoidal rule over the AVAC frame times. The resulting ledger contains injected water volume, both depth-momentum integrals, and both physical water-momentum components after multiplication by ρ_w .

The ledger separates a transfer error from a propagation error. Its volume and momentum can be compared directly with the positive avalanche flux implied by Eqs. (35)–(38), while the WAVE state can be integrated independently after injection. These two balances form the first part of the coupling verification in Sect. 4.3.

2.4.6 Physical scope of the sequential coupling

The transfer is one-way and offline. AVAC is completed before WAVE starts, so water pressure and wave motion do not feed back on avalanche runout. The AVAC state is not removed or altered after crossing the polygon; its inward flux is used only to construct an equivalent water source. Consequently, the method does not track submerged granular material or explicitly solve buoyancy, hydrodynamic drag, entrainment, mixing, compaction, melting, or air displacement.

Within WAVE, the converted material is inserted into the polygon-mask shoreline source-target cells (which can include the zero-depth stabilization rim) and thereafter behaves as water governed by Eqs. (29)–(31). This construction is less complete than two-phase or multilayer impact models, but is computationally transparent and compatible with a geospatial hazard workflow. Its scientific claim is correspondingly narrow: it is a conservative depth-integrated shoreline forcing when $\varepsilon_T = 1$, followed by hydrostatic long-wave propagation. The forthcoming laboratory comparison must determine the useful range of ε_T and the conditions under which this reduction is adequate.

2.5 QGIS integration, managed runtimes, and outputs

AVAC4QGIS embeds the model chain in QGIS rather than requiring the user to construct GeoClaw data files manually. The plugin reads a selected DEM band, transforms release and lake polygons into its coordinate reference system, and provides previews of the release depth, initial snow surface, lake depth, and stabilized shoreline before execution. Scientific controls remain explicit: the interface writes the selected rheology,

Planned figure: internal shoreline transfer

Show a lake polygon rasterized on the WAVE grid, inward face normals, sampled AVAC (h, hu, hv), the positive normal flux filter, the source triplet (Q, Qu, Qv), temporal interpolation, and the receiving WAVE cells.

Figure 2: Working placeholder for internal shoreline transfer.

release corrections, grid spacing, duration, output frequency, wet-dry tolerance, friction, CFL, and coupling coefficient into human-readable run configuration files.

Preparation checks a valid coordinate reference system and positive square horizontal cell spacing. Release and lake polygons are transformed into the selected DEM coordinate reference system. Users must ensure that this CRS has suitable metric horizontal units, that raster orientation and sampling are appropriate for a Cartesian calculation, and that DEM and water-level vertical datums are compatible; these properties cannot in general be inferred from the layer metadata. The WAVE grid must cover the AVAC domain with an integer number of cells, and its terrain/bathymetry DEM must cover that domain at an equal or integer-finer source resolution. A coupled run cannot be prepared from an unfinished AVAC calculation or from a lake shoreline entirely outside AVAC coverage. Each calculation is materialized in a new run directory containing copies of its geospatial inputs, configuration, generated solver files, status metadata, and outputs. Thus later changes to a QGIS layer do not silently rewrite the provenance of an existing result.

The numerical executables and their required native libraries are distributed as managed, platform-specific runtimes. At installation the plugin selects the runtime for the operating-system/architecture pair, checks its manifest and cryptographic hashes, and installs it atomically in the QGIS profile. A user therefore does not need a Fortran compiler, Make, a Python Clawpack installation, or a separately configured Conda environment. The packaging system contains target selection for macOS on Apple Silicon, Windows on AMD64, and Linux on x86_64; the set of binaries formally released and tested for the paper is reported separately in the reproducibility matrix rather than inferred from this selection logic.

After a run, the plugin reads GeoClaw patch output into a shared QGIS results model. Standard temporal AVAC layers are depth, terrain-tangent speed, snow-surface elevation, and the derived kinetic-pressure proxy

$$p_k = \frac{1}{2} \rho_a U_{a,s}^2 / 1000 \quad (\text{kPa}). \quad (40)$$

It is neither a solver pressure nor a basal-stress, impact-pressure, or constitutive prediction. The raw output frames retain the conserved momenta, but the ordinary QGIS temporal layer choices do not expose a momentum map. Summary products include maximum depth, speed, and kinetic pressure, plus arrival and time-of-maximum fields where available; the last saved temporal frame supplies the final state rather than a separate deposition product. The temporal reader currently masks speed and kinetic pressure below a fixed depth of 0.05 m and uses an abrupt momentum/depth ratio. It can therefore disagree with a user-selected h_v or the smooth fixed-grid maximum diagnostic. WAVE results add lake-volume and shoreline-source diagnostics, while coupled scenarios retain a link to the source AVAC run and its transfer ledger. AVAC and WAVE use the same civil-time origin in QGIS, so their layers can be compared on one timeline even though the solvers execute sequentially.

These controls make the workflow “plug and play” in the operational sense: installation and routine execution do not require solver compilation or command-line preprocessing. They do not remove the need for scientific judgement. Release geometry, DEM quality, rheological parameters, water level, resolution, boundary distance, and the coupling efficiency remain modelling choices whose consequences must be tested and documented.

3 Evaluation strategy

3.1 Verification, validation, intercomparison, and demonstration

In this paper, *verification* assesses whether the numerical implementation solves the intended mathematical problem, whereas *validation* assesses whether the physical model represents observed avalanche behavior. The ISeeSnow cases are treated as an *intercomparison*: participating-model output is not assumed to be truth. A coupled application without observational data is called a *demonstration* rather than a validation.

3.2 Evidence matrix

Table 1: Evidence planned for each principal model claim.

Claim	Primary evidence	Location
AVAC solves its intended equations	Kerswell horizontal Coulomb spreading, sloping-bed Coulomb flow, and local curvature normal-load source solutions	Sect. 4.1
WAVE preserves equilibrium and moving wet/dry solutions	Lake at rest and Thacker paraboloid oscillation	Sect. 4.2
The transfer operator supplies the intended volume and momentum	Prescribed shoreline-flux ledger, closed-domain balance, grid and AMR invariance	Sect. 4.3
AVAC represents observed avalanche behavior	Laboratory or field data to be selected	Sect. 5
AVAC behaves consistently under a shared model protocol	Three official ISeeSnow cases with prescribed physical parameters; numerical choices assessed using PFT	Sect. 6
The complete workflow is fit for its stated scope	Planned coupling experiment and cross-platform reproducibility checks	Sects. 7 and 8

4 Verification

4.1 AVAC verification

Checkpoint of the illustrated verification. The two Coulomb panels and the water-limit appendix retain the supplied pre-1.0.0 calculations, identified in their companion JSON by AVAC executable SHA-256 beginning 6e85bee646c3. The values below are synchronized with that JSON rather than mixed with later Windows runs. These illustrations are not a new full AMR verification of version 1.0.0. For the final Voellmy stopping correction alone, paired native controls preserved uniform sloping Coulomb and frictionless water fields exactly; horizontal Kerswell AMR depth differed by at most 1.15×10^{-15} m, with unchanged reported analytical metrics. Earlier changes to the static-interface guard did alter some granular flat states. Thus the narrow stopping-fix regression is not evidence that every change since the illustrated checkpoint leaves all flat trajectories unchanged. No new simulations were run for this documentation and version-number update.

The supplied manuscript illustrates AVAC verification with two finite-volume Coulomb dam-break problems derived from the exact shallow-layer solution of Kerswell (2005) and with closed local tests of the curvature-dependent normal-load source. The horizontal case isolates pressure-driven spreading, basal resistance, wet-dry propagation, and natural arrest. The inclined case retains the same solution structure while adding the downslope gravity source and the reduction of normal stress with slope. Taken together, the tests exercise the parts of the momentum balance that distinguish AVAC from a water-only shallow-flow solver. A third, frictionless water-limit test is reported in Appendix A.

Each nominally one-dimensional problem was solved by the two-dimensional AVAC executable in a five-cell-wide strip with wall boundaries in the transverse direction. Three AMR levels were used, with two 4:1 longitudinal refinements and no transverse refinement. The horizontal case therefore progressed from a 40 mm background mesh to 2.5 mm near the front and rear boundaries; the corresponding spacings for the inclined case were 30 mm and 1.875 mm. Narrow, overlapping space-time refinement corridors followed the independently evaluated analytical boundary paths. These paths controlled mesh allocation only: neither the conserved state nor the numerical flux received analytical information. Thus the fine mesh remained concentrated near both moving boundaries while the rest of the domain stayed coarse.

For the horizontal case, the numerical front was obtained from the rightmost cell whose depth exceeded the solver dry tolerance. For the inclined case, the energy-resolved support archived by the validation notebook was used so that the reported front represents dynamically resolved material. Its post-processing threshold is $\sqrt{h_a}|U_h| > 5 \times 10^{-5} \text{ m}^{3/2} \text{ s}^{-1}$; it is independent of the solver configuration and is not a velocity ceiling. The rear boundary was the downstream edge of material whose depth remained within 0.1 % of H_0 ; this finite tolerance prevents harmless AMR interpolation perturbations from being classified as motion. Boundary errors are reported as root-mean-square errors (RMSEs) over the corresponding analytical moving interval. Depth and velocity were sampled along the strip centreline, and conservation was evaluated from the integral of flow depth per unit transverse width. The notebook for each test records the complete controls, the solver checksum, the achieved AMR level, centreline fields, and boundary time series; consequently, Fig. 3 can be identified by the supplied diagnostic summary. The rendered figures and summaries are retained; regenerating any unavailable full native history requires rerunning the corresponding source and controls.

4.1.1 Kerswell horizontal Coulomb dam break

The initial condition was a static column of depth $H_0 = 1$ m on $-10 \text{ m} \leq x \leq 0$, with a dry bed downstream and a Coulomb coefficient $\mu = 0.1$. The upstream boundary was a wall and the downstream boundary was extrapolating. With

$$X_0 = \frac{H_0}{\mu} = 10 \text{ m}, \quad T_0 = \frac{1}{\mu} \sqrt{\frac{H_0}{g}} = 3.19 \text{ s}, \quad \tau = \frac{t}{T_0}, \quad (41)$$

the analytical front advances according to $x_f/X_0 = 2\tau - \tau^2/2$ until it arrests at $\tau = 2$ and $x_f/X_0 = 2$; the rear boundary arrests earlier, at $\tau = 1.529654$ and $x_b/X_0 = -0.721577$ (Kerswell, 2005).

AVAC reproduced both boundary trajectories and the interior depth and velocity profiles (Fig. 3a–c). During motion, the front and rear RMSEs were 0.1585 m and 0.003 11 m, respectively. At 10 s, the simulated front was at 19.831 m, compared with the analytical 20 m, and the maximum remaining speed was $1.69 \times 10^{-4} \text{ m s}^{-1}$. The depth integral was initially 10 m^2 per metre of transverse width and varied by $6.15 \times 10^{-4} \text{ m}^2$, or $6.15 \times 10^{-3} \%$. The agreement in the arrested state is especially relevant because no prescribed stopping time or fitted velocity threshold was used: arrest follows from the Coulomb and static-yield terms described in Sect. 2.2.

4.1.2 Coulomb dam break on a sloping bed

The second calculation used the same released depth, but placed the column on a planar bed descending at $\theta = 5^\circ$ and increased the Coulomb coefficient to $\mu = 0.2$. In the downslope coordinate, gravity and basal resistance combine into an effective friction coefficient

$$\mu_{\text{eff}} = \mu - \tan \theta = 0.1125, \quad (42)$$

while the pressure-wave speed is based on $g \cos \theta$. Consequently, the horizontal analytical solution can be reused with $X_0 = H_0/\mu_{\text{eff}} = 8.89 \text{ m}$ and $T_0 = \sqrt{H_0/(g \cos \theta)}/\mu_{\text{eff}} = 2.84 \text{ s}$. This equivalence provides a direct test of the balance between the topographic source and Coulomb resistance implemented in AVAC, rather than merely a second wet-dry propagation test.

The resulting dimensionless trajectories and profiles closely followed the same analytical curves as the horizontal case (Fig. 3d–f). The moving-front and rear RMSEs were 0.1232 m and 0.0342 m, respectively. At 6 s, the illustrated run placed the resolved front at 17.6253 m, compared with the analytical 17.776 m; the rear positions were -6.5209 m and -6.4082 m . The maximum resolved speed had decreased to $1.14 \times 10^{-3} \text{ m s}^{-1}$. Over the calculation, the 9.99 m^2 initial depth integral varied by $9.07 \times 10^{-4} \text{ m}^2$, corresponding to $9.08 \times 10^{-3} \%$. These are the supplied historical diagnostic definitions, not a recomputation using the current release’s controls.

The two Coulomb tests are complementary. The horizontal calculation shows that AVAC propagates and arrests a finite granular mass without gravitational driving along the bed, whereas the inclined calculation shows that the same behavior is recovered after the driving and resisting accelerations are combined. Their common dimensionless solution therefore links the observed agreement to the implementation of the AVAC source terms, rather than to a case-specific parameter adjustment.

4.1.3 Local curvature normal-load source

The curvature-dependant normal-load term was checked independently of the finite-volume flux by compiling the same Fortran rheology module used by AVAC and comparing its scalar frozen-coefficient update with closed solutions. A material element with initial map speed 8 m s^{-1} and $\mu = 0.30$ was evaluated for 4 s at the nadir of planar, concave-circular, and convex-circular tracks. The circular radius was 40 m, giving $\kappa_v = \pm 0.025 \text{ m}^{-1}$, and the analytical equation was

$$\frac{dU_h}{dt} = -\mu \left(g + \kappa_v U_h^2 \right). \quad (43)$$

The maximum absolute speed errors were below $3.2 \times 10^{-15} \text{ m s}^{-1}$. Repeating a 2 s update as eight equal source substeps changed the result only at machine precision.

A zero-force test of the isolated *scalar basal routine* leaves its input map speed unchanged. It must not be interpreted as the behaviour of the full current source step, which first performs the separate terrain-tangent transport. The production regression tests additionally exercise affine-bed identity, tangent-speed preservation, the differential connection limit, rotated two-dimensional geometry, physical boundaries

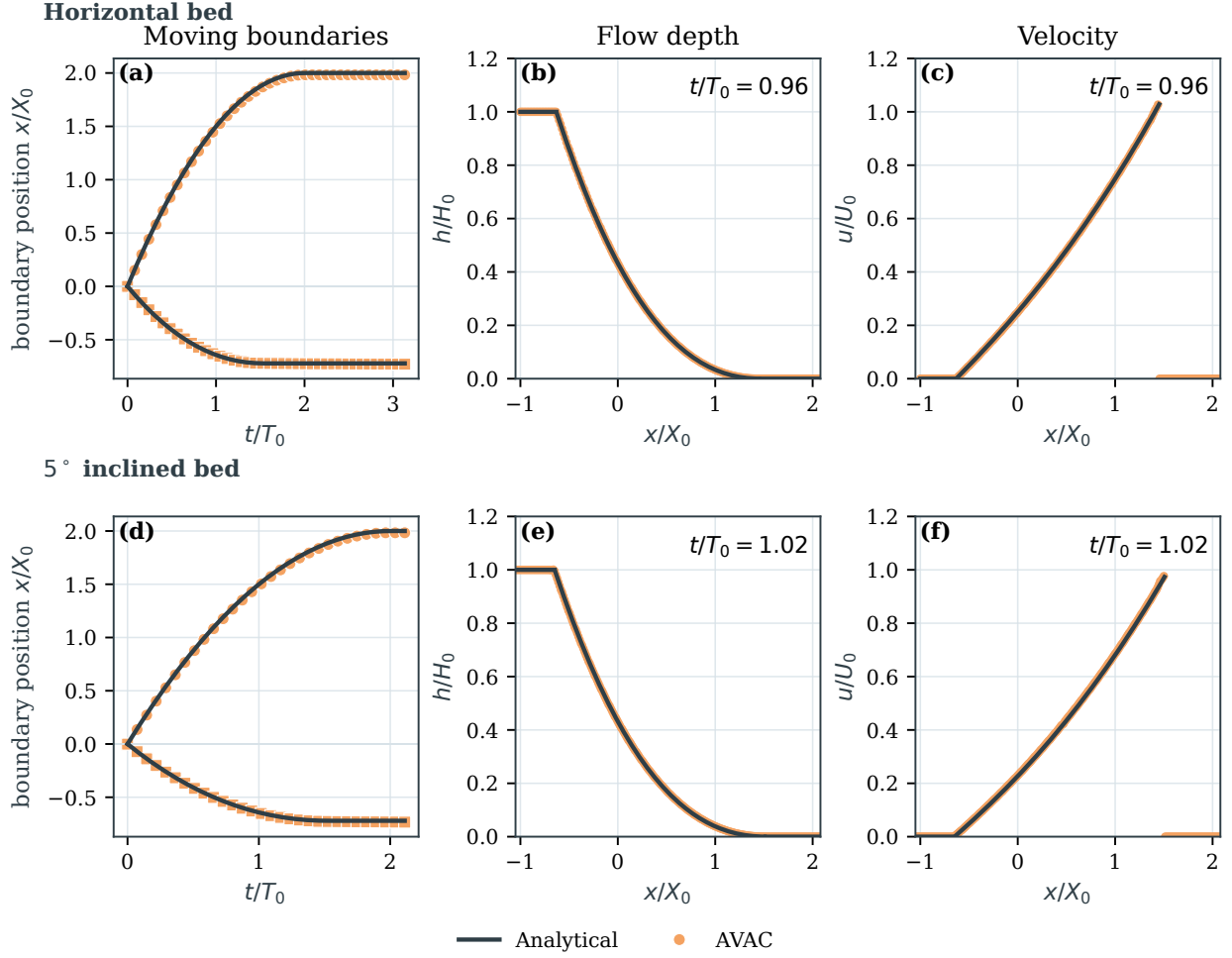


Figure 3: AVAC verification against the Coulomb dam-break solution of [Kerswell \(2005\)](#). The upper row shows the horizontal-bed case and the lower row the 5° inclined-bed case. Panels (a,d) compare the front (circles) and rear (squares) trajectories; panels (b,e) compare flow depth; and panels (c,f) compare downslope velocity at $t/T_0 \simeq 1$. Dark lines denote the analytical solution and orange symbols denote AVAC centreline results. AMR provides finest longitudinal spacings of 2.5 mm and 1.875 mm in the two rows, respectively. These are the supplied pre-1.0.0 verification fields (executable prefix 6e85bee646c3), not a new release-wide AMR run.

and patch partitioning. These checks test distinct operators rather than replace a resolved curved-DEM trajectory or establish convergence of their split composition.

The circular-track values above verify the normal-load term for frozen local geometry. Their convex state remains below the loss-of-contact threshold in Eq. (17); that particular example does not exercise the branch switch. Compiled source-stopping tests cover the corrected Voellmy and cohesive-Voellmy zero-result handling, including near-dry states and unchanged Coulomb/water controls. The released test sources retain these separate scopes explicitly.

4.2 WAVE verification

The WAVE figures and their numerical summaries below are retained from the supplied manuscript. They were not rerun for version 1.0.0; the Windows WAVE executable is unchanged from the preceding release. They remain component-verification evidence at their recorded configurations, not a new cross-platform equivalence experiment.

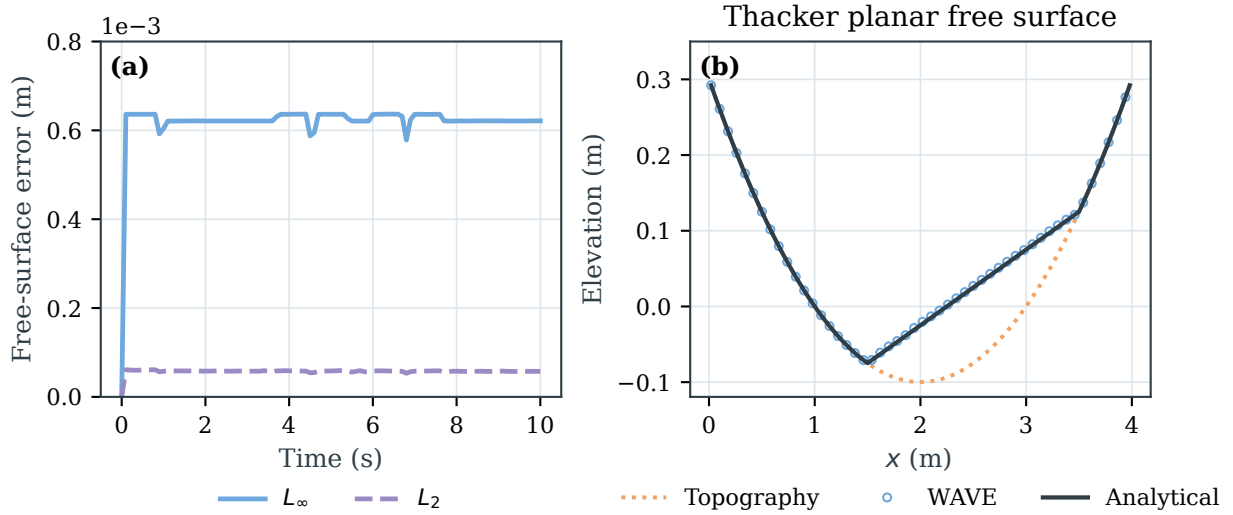


Figure 4: WAVE verification. (a) L_∞ and L_2 free-surface errors for the lake-at-rest test over a smooth bump. (b) Centreline free surface after the third period of the planar Thacker oscillation in a parabolic basin. Black is the analytical solution, blue open circles are WAVE, and orange dotted line is the topography. Legends are placed outside the plotting regions.

4.2.1 Lake at rest

The first WAVE test asks whether the lake initialization and the discretized bathymetric source remain balanced before any avalanche source is applied. The bed contains the smooth bump used by the Baines steady-flow test, while the initial free surface is horizontal at 2 m. The 25 m by 1 m domain is resolved with 0.05 m cells, the longitudinal boundaries are closed, and the transverse boundaries are periodic. The coupling file contains no active source. Consequently, the exact solution is a lake at rest for the complete 10 s calculation.

The free-surface errors rose during the initial numerical adjustment and then remained nearly constant (Fig. 4a). The largest L_∞ and L_2 errors over all saved times were 6.37×10^{-4} m and 6.11×10^{-5} m, respectively. Thus the topographic balance introduces a submillimetric disturbance, but no growing wave. This test is especially relevant to the coupled workflow: it places an upper bound on the background signal against which an avalanche-generated impulse must be distinguished.

4.2.2 Thacker planar oscillation in a paraboloid

The still-water calculation does not exercise shoreline motion. We therefore use the exact planar free-surface oscillation in a parabolic basin derived by Thacker (1981), with reference values generated independently through the SWASHES collection (Delestre et al., 2013). A 4 m square basin is represented by a 100×100 grid, giving 0.04 m cells, and wall conditions are imposed on all four sides. The calculation covers three complete oscillation periods, so that phase, amplitude, and repeated wetting and drying all contribute to the final error.

Figure 4b compares the WAVE and analytical centreline free surfaces over the parabolic topography after the third period. The WAVE centreline depth follows the analytical wet support and profile closely; the depth RMSE is 1.27×10^{-3} m, and the maximum absolute error is 2.84×10^{-3} m. Together with the lake-at-rest result, this shows that WAVE both maintains a nominally stationary lake and transports a moving shoreline through a genuinely two-dimensional basin without a systematic phase shift visible at the reported resolution.

The remaining analytical profiles and numerical diagnostics are collected in Appendix B. They broaden the evidence to steady hydraulic transitions, shocks, friction, dry fronts, variable channel width, AMR accuracy,

and parallel reproducibility without interrupting the stationary-to-moving-lake progression used here.

4.3 Coupling verification

The component tests above verify the two solvers separately, but not the transfer between them. We therefore prescribed a spatially uniform synthetic AVAC state along a straight shoreline of length $L = 0.8$ m. Over $T = 1$ s, its depth was $h_a = 0.1$ m and its velocity was

$$(u_a, v_a) = A(t)(0.02, 0.01) \text{ m s}^{-1}, \quad A(t) = \sin(\pi t/T) \left(1 + \frac{t}{4T}\right). \quad (44)$$

The pulse is zero at both ends and has closed-form volume and momentum integrals. It was written as native AVAC frames, sampled by the production shoreline converter, serialized in the conservative source format, read by the WAVE Fortran source module, and applied by the current WAVE executable. Thus the test exercises the complete numerical transfer rather than only the Python rate calculation.

For a uniform active shoreline, the expected transferred volume, depth-integrated momentum, and physical water momentum are

$$V_w = \beta L \int_0^T h_a u_n dt, \quad (45)$$

$$I_x = \beta L \int_0^T h_a u_n u_a dt, \quad P_x = \rho_w I_x, \quad (46)$$

$$I_y = \beta L \int_0^T h_a u_n v_a dt, \quad P_y = \rho_w I_y, \quad (47)$$

where u_n is the velocity component directed into the lake and $\beta = 1$ for this verification. Equation (44) gives $V_w = 1.14592 \times 10^{-3} \text{ m}^3$, $I_x = 2.02827 \times 10^{-5} \text{ m}^4 \text{ s}^{-1}$, and $I_y = 1.01413 \times 10^{-5} \text{ m}^4 \text{ s}^{-1}$. The WAVE domain was a frictionless, periodic 2 m by 1 m basin with an initial uniform depth of 0.5 m. Periodicity removes boundary losses, so changes in the three globally integrated conserved variables can be compared directly with the written source ledger.

The prescribed and serialized volume-rate histories are visually coincident (Fig. 5a). WAVE was run on uniform 0.10 m and 0.05 m grids and with two AMR levels from a 0.10 m base grid to a 0.01 m finest grid. Integration of the non-overlapping native control volumes gave a maximum absolute relative closure error of 5.61×10^{-5} across volume and both momentum components (Fig. 5b). The AMR run reached level 2 and reduced the largest closure error to 2.95×10^{-7} ; the result therefore confirms that the format-2 total rates remain conservative when the receiving cell area changes.

These tests verify numerical transfer and conservation, but they do not validate the physical assumption that an entering avalanche can be replaced by an equivalent one-way water source. That question requires the planned laboratory experiment in Sect. 7; the distinction between numerical coupling verification and physical coupling validation is maintained throughout this paper.

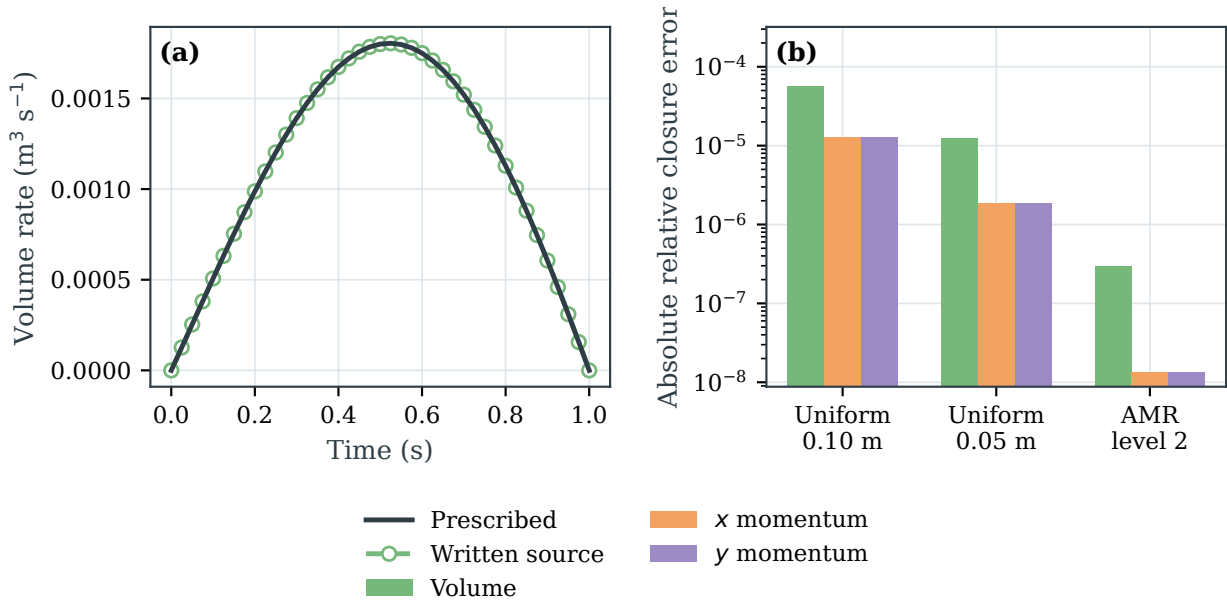


Figure 5: Numerical verification of the AVAC-to-WAVE coupling. (a) Prescribed and serialized volume-rate histories. (b) Absolute relative closure errors for volume and both depth-momentum components on two uniform grids and a level-2 AMR grid. The complete path includes native AVAC frames, the production converter and source file, the Fortran source update, and native WAVE control volumes.

5 AVAC validation against observations

Before drafting this section, identify a comparable avalanche-model validation paper with accessible release geometry, terrain, material parameters, and observations of runout, front motion, velocity, or deposit. Prefer an open dataset and a case that permits either independently prescribed parameters or calibration on one event followed by validation on another. A single case fitted and assessed with the same outputs must be described as calibration or evaluation, not independent validation.

Once data are selected, use the following subsection structure:

1. observational or experimental case and uncertainty;
2. numerical setup and parameter-selection strategy;
3. runout, front, velocity, deposit, and volume results;
4. separation of input uncertainty from model-structure limitations.

6 ISeeSnow avalanche-model intercomparison

6.1 Protocol and common metrics

The ISeeSnow v1.0 exercise provides common idealized and real topographies, release areas, rheological parameters, and submitted peak fields from several depth-integrated avalanche models (Wirbel et al., 2026). We ran AVAC on the supplied 5 m grid without retuning the prescribed physical parameters. Numerical limiter and shallow-flow controls were assessed using peer PFT comparisons, as distinguished below. The three calculations are the idealized Voellmy case ($\mu = 0.4$, $\xi = 2000 \text{ m s}^{-2}$), the real-topography Voellmy case ($\mu = 0.2$ with the same ξ), and the idealized Coulomb-only case ($\mu = 0.4$ and no turbulent resistance). In all three cases, the prescribed terrain-normal release thickness is 1.5 m.

The two idealized cases share the same inclined plane, variable-width ravine, release area, and flat foreland, while the real-terrain case uses a distinct mountain catchment and release polygon (Fig. 6). Thus the first and third calculations contrast Voellmy and Coulomb-only resistance on a common geometry, using the model-specific numerical controls stated below, whereas the first and second retain Voellmy resistance while changing both terrain and the prescribed Coulomb coefficient.

Because ISeeSnow prescribes release thickness normal to the terrain whereas AVAC advances vertical map-cell depth, initialization applies the inverse geometric conversion

$$h_a(x, y, 0) = \frac{h_n(x, y, 0)}{\cos \phi(x, y)}. \quad (48)$$

The prescribed normal release volume is therefore represented consistently on sloping cells before the calculation begins.

AVAC stores vertical depth and horizontal map-coordinate momentum, whereas ISeeSnow prescribes terrain-normal peak flow thickness (PFT). We therefore convert vertical peak depth back to terrain-normal PFT cell by cell as $\text{PFT}_n = \text{PFT}_h \cos \phi$. PFV is the physical terrain-tangent magnitude consistent with Eq. (8), $[u_a^2 + v_a^2 + (u_a \partial_x B + v_a \partial_y B)^2]^{1/2}$, evaluated at every solver step before taking the temporal maximum. The release state at $t = 0$ is included explicitly in the PFT maximum, and subsequent maxima are obtained from the native peak-output calculation. PFV is zero for vertical depth at or below 0.05 m, uses the smooth diagnostic transition described in Sect. 2.2 between 0.05 m and 0.20 m, and is the exact terrain-tangent velocity at larger depth. This follows the model-specific minimum-depth criteria documented by ISeeSnow. The AVAC run configuration sets the shared GeoClaw speed limit to 10^{99} m s^{-1} , so neither the evolving state nor the reported PFV is clipped by a finite velocity ceiling.

Two peer populations serve different questions. The scalar overview uses the published Table C1 core group: 11 idealized-Voellmy, 10 real-terrain, and 11 Coulomb submissions. It compares thalweg runout, maximum PFT and maximum PFV. Runout is the furthest thalweg coordinate with normal PFT exceeding 0.5 m; it is not the area of a 0.01 m support mask. The off-scale Coulomb PFV of r.avafLOW is retained in all scalar statistics and annotated in the figure, not discarded.

Spatial PFT comparison, the primary numerical-selection criterion, instead uses only rasters with exactly matching dimensions, cell size and sample coordinates: 9, 8 and 8 peers for the three cases. No field is shifted or resampled to improve agreement. Its dimensionless score is

$$E_{\text{PFT}} = \text{median}_i \left[\frac{\text{RMSE}(\text{PFT}_{\text{AVAC}}, \text{PFT}_i)}{\text{median}_{j \neq i} \text{RMSE}(\text{PFT}_i, \text{PFT}_j)} \right], \quad (49)$$

using full-grid errors and the recorded common nodata treatment. It is not the unnormalized median RMSE in metres, nor an observational accuracy score. The broader scalar population must not be confused with this spatially aligned population.

The current comparison uses the accepted executable whose SHA-256 begins 55a8c4baf0cc; version 1.0.0 contains the same binary. All cases use a uniform 5 m grid, second-order spatial updating, a 1200 s horizon and 10 s saved-state intervals. The two Voellmy cases use Van Leer and target CFL 0.50; Coulomb

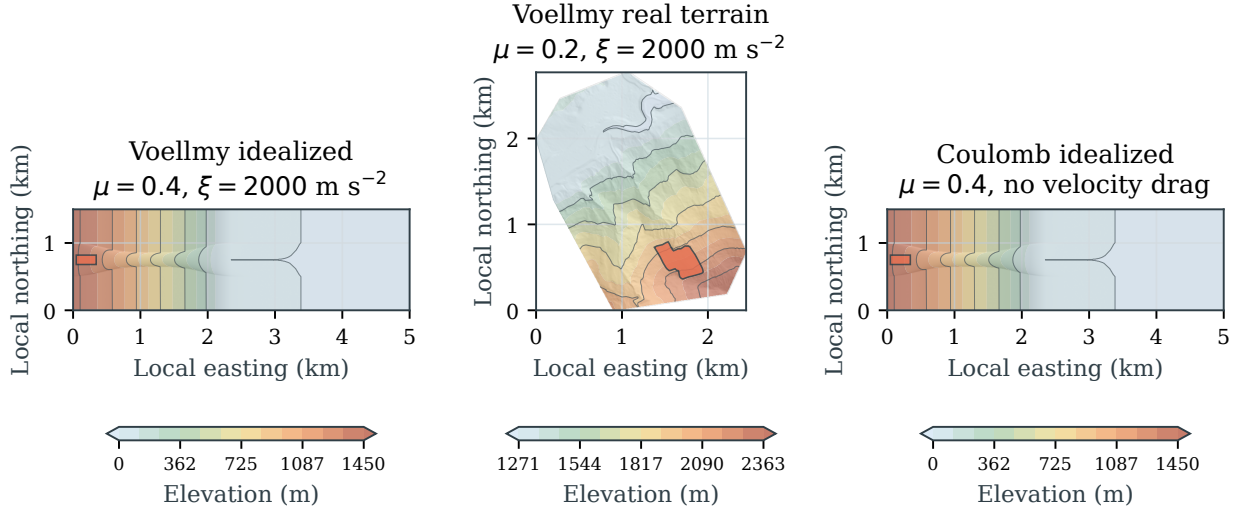


Figure 6: The three AVAC ISeeSnow configurations, drawn from the supplied benchmark DEMs and release polygons. (a) Idealized topography with Voellmy resistance, (b) real topography with Voellmy resistance, and (c) the same idealized topography with Coulomb resistance only. Red polygons denote the common 1.5 m terrain-normal release thickness. The maps are an original rendering of the inputs used in Wirbel et al. (2026).

uses Minmod and target CFL 0.25. The correction-flux depth controls are respectively 0.10 m and 0.05 m; the solver dry tolerance is 1×10^{-4} m and the rejection ceiling is CFL 1.0. These are explicit benchmark controls, not an assertion that Van Leer or CFL 0.25 is the general plugin default. Physical friction coefficients and curvature treatment were not fitted to peer output, but the numerical choices were assessed using PFT agreement; this is therefore not a blind validation exercise.

6.2 Current fields and comparison with peers

Table 2: Accepted ISeeSnow results carried into version 1.0.0. The PFT score uses aligned spatial peers; the figure uses Table C1 scalars.

Case	Runout (km)	Peak PFT (m)	Peak PFV (m/s)	E_{PFT}
Idealized Voellmy	2.31005	8.53216	39.6255	0.865517
Real-terrain Voellmy	2.09712	21.03670	67.6648	0.907803
Idealized Coulomb	3.15004	7.21983	110.9650	0.911514

For idealized Voellmy, runout and peak PFT lie within the scalar peer interquartile intervals, 2.305 km to 2.335 km and 7.635 m to 9.285 m. Peak PFV is slightly above the upper quartile of 39.285 m s^{-1} , but below the peer maximum of 41.55 m s^{-1} . For real terrain, runout and peak PFT are also within their interquartile intervals. PFV lies above the upper quartile of 52.465 m s^{-1} , within the full 40.87 m s^{-1} to 80.23 m s^{-1} range.

The Coulomb runout lies within the peer interquartile interval 3.134 34 km to 3.226 52 km. Peak PFT is slightly above its upper quartile of 7.125 m, while remaining within the 3.95 m to 9.97 m range. PFV is likewise just above its upper quartile of $110.7155 \text{ m s}^{-1}$. Thus the accepted Coulomb runout is comparable with the central peer population, but the deposit crest is not identical to a peer median. The peak normal thickness is 7.219 83 m; its cell retains 7.189 42 m at the horizon with zero momentum. The recorded direct-interface guard correction did not alter this Coulomb trajectory relative to the preceding experimental V1 cohort: the complete saved native histories and final PFT/PFV fields match. The crest is not attributed to that separately demonstrated guard defect.

The peer products are maxima over time, not observed final-deposit histories. Different peer crests occur

at different coordinates, so a pointwise median can be lower than the median of their individual maxima. Neither a peer envelope nor the scalar agreement in Table 2 identifies a unique physically correct deposit.

6.3 Conservation, practical rest and remaining sensitivity

The native relative volume changes $(V_f - V_0)/V_0$ are 1.6463×10^{-9} , 6.0960×10^{-10} and -6.4338×10^{-10} for idealized Voellmy, real-terrain Voellmy and Coulomb. Practical-rest confirmation occurs at 590 s, 520 s and 100 s, respectively, and remains satisfied at subsequent saved checks through the horizon. This criterion requires less than 1% of the release volume to be moving faster than 0.01 m s^{-1} in the resolved depth band above 0.05 m, for at least three saved times. It is not zero velocity in every wet cell. Final resolved moving volumes are approximately 70.09 m^3 , 231.35 m^3 and zero; motion below the reporting-depth threshold is audited separately.

The correction that honors Voellmy's exact zero source result removed the reproduced extremely large near-dry post-source states. Both full Voellmy observation-only audits reproduced all 484 saved native artifacts and the PFT/PFV fields; no invalid accepted post-source states were observed. However, real terrain still has a raw all-step terrain-tangent maximum of 90.97 m s^{-1} in the 0.05 m to 0.10 m depth band, distinct from its filtered PFV of 67.66 m s^{-1} . Neither saved frames nor filtered peak fields bound every intermediate split state.

A same-source Voellmy target-CFL refinement from 0.50 to 0.25 reduced the real-terrain thin-layer pulse and improved its normalized PFT score by about 1.52%, but worsened the idealized PFT score by about 1.84%. PFV and local fields changed non-uniformly. Accordingly, the existing controls are retained; two timestep levels do not establish a convergence rate or universal stability. The improved PFT comparisons, the separate flat-control evidence and these remaining sensitivities are reported together rather than treating peer agreement as a substitute for numerical verification.

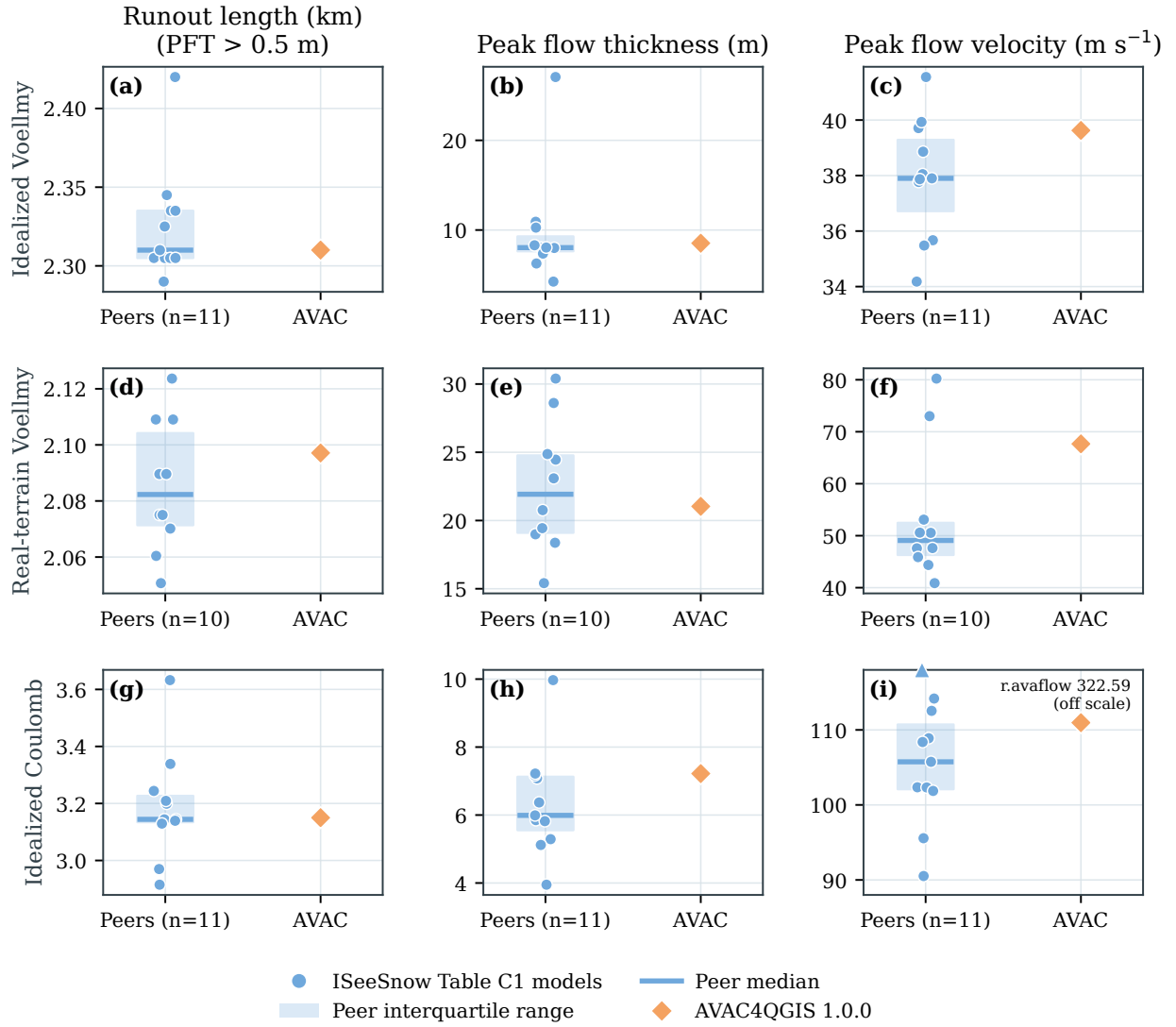


Figure 7: Version 1.0.0 in the ISeeSnow Table C1 core-model ensemble. Rows show idealized Voellmy, real-terrain Voellmy and idealized Coulomb; columns show thalweg runout for PFT greater than 0.5 m, peak PFT and peak PFV. Blue points, bands and lines are peer scalars, interquartile ranges and medians; orange diamonds are the accepted AVAC fields. All 11/10/11 peer values are retained. The off-scale Coulomb velocity is annotated and included in the statistics. This is a replot of authenticated completed-run metrics, not a newly executed validation experiment.

7 Coupled avalanche–impulse-wave validation experiment

Write after the planned laboratory experiment is completed. Planned subsections:

1. experimental geometry, material properties, release, water depth, and measurement uncertainty;
2. AVAC motion and shoreline-arrival state;
3. transferred volume and momentum ledger;
4. WAVE gauge histories, amplitudes, wavelengths, and runup;
5. sensitivity to β , grid spacing, shoreline representation, and AVAC output interval;
6. discrepancies attributable to hydrostatic and one-way coupling assumptions.

Reserve the terms “coupling validation” and “validated coupled model” until the experiment and acceptance metrics justify them.

8 Discussion

Draft once all evidence is available. Planned subsections:

1. what component verification, AVAC validation, ISeeSnow, and the coupling experiment establish;
2. physical meaning and limitations of the one-way shoreline transfer;
3. hydrostatic/non-dispersive limitations of WAVE;
4. grid, shoreline, and temporal-sampling sensitivities;
5. accessibility and reproducibility through QGIS and managed runtimes;
6. future two-way, multilayer, or dispersive developments.

The multiplatform claim must be supported by a common-case scientific- equivalence test on every released platform, with declared numerical tolerances.

9 Conclusions

Four short paragraphs: the model-system contribution; evidence for AVAC and WAVE; evidence from ISeeSnow and the coupling experiment; and the final scope-of-use statement.

10 Code and data availability

The source and Windows plugin are released together as version 1.0.0 at <https://github.com/cgotelli/AVAC-QGIS/releases/tag/v1.0.0>; the integrated development branch is `main`. The editable manuscript, bibliography, figures and replot script are stored under `docs/article`. The release also provides the compiled methods PDF, the installable Windows AMD64 plugin ZIP, its SHA-256 checksum and a runtime manifest. The main source tree retains the validation drivers and tests; third-party components and their supplied licences are described in `THIRD_PARTY_NOTICES.md`.

The ISeeSnow input geometry and peer submissions originate from the common protocol of Wirbel et al. (2026). The updated scalar figure retains the archived input-data hash, exact AVAC solver identity, Table C1 values and portable replot data. It can be redrawn without a solver run. Selected final PFT/PFV fields, controls, diagnostic reports and earlier worktree patches were preserved in compact local evidence archives during release preparation; bulk intermediate native histories were removed. Historical audit reports remain records of the completed checks, but repeating their full byte-level audits requires regenerating those histories. This repository deposit is not claimed to have a DOI or to be a complete permanent archive of every intermediate field. Planned observational and laboratory-validation data are not represented as available results.

A Additional AVAC verification

As an end-member test, AVAC was also configured with zero basal friction and no avalanche rheology and compared with the exact finite-volume dam-break solution of [Ancy et al. \(2008\)](#). A triangular prism of water with $H_0 = 1$ m was released on a frictionless 10° planar bed; its initial depth was $h(x, 0) = H_0(1 - x/x_b)$ for $x_b \leq x \leq 0$, where $x_b = -H_0/\tan(10^\circ)$. The calculation used the same three-level, front-focused AMR procedure as the main AVAC tests on a domain extending from $x = -10$ m to $x = 60$ m, progressing from a 40 mm background spacing to 2.5 mm near the head and tail. This case is placed in the appendix because it verifies the frictionless shallow-water limit shared with GeoClaw, rather than avalanche-specific resistance.

A constant 10 m vertical datum was retained in the planar-bed input used for this benchmark. The present AVAC configuration sets GeoClaw’s reference sea level to -10^4 m, safely below the computational terrain, so the translation is no longer required to keep newly refined dry cells dry. It is nevertheless immaterial to the comparison because a uniform vertical translation changes neither bed gradients, water depth, nor the analytical solution.

The extended domain retained both moving boundaries throughout the complete 5 s comparison. Over that interval, the front and rear RMSEs were 0.062 66 m and 0.080 56 m, respectively. The maximum initial-depth discrepancy on the uniform diagnostic grid was 3.31 mm. The initial depth integral, $2.835\,64\text{ m}^2$ per metre of transverse width, differed from the exact value by $1.31 \times 10^{-7}\text{ m}^2$ and varied by $4.15 \times 10^{-4}\text{ m}^2$ over the full simulation, or 0.0146 %. These errors include the interpolation from the front-focused AMR hierarchy to the one-row diagnostic grid.

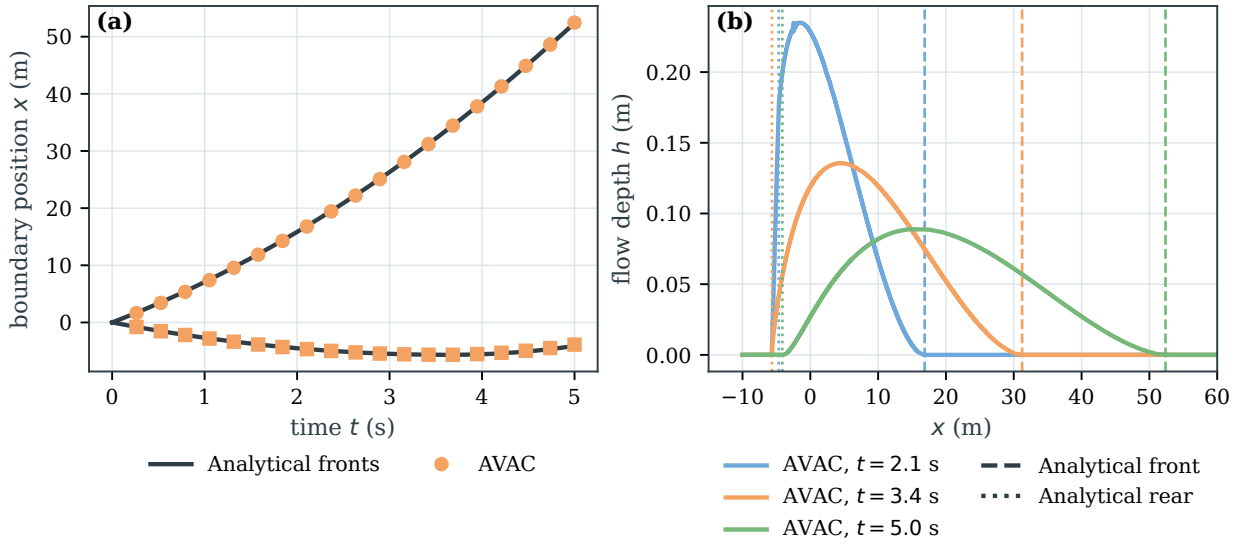


Figure 8: Frictionless AVAC water-limit verification against [Ancy et al. \(2008\)](#). (a) Analytical and numerical positions of the advancing front and rear boundary over the complete 5 s simulation. (b) AVAC depth profiles at approximately 2.1 s, 3.4 s, and 5 s; dashed and dotted vertical lines mark the corresponding analytical front and rear positions. The profiles are produced from saved centreline fields and are not solver inputs. This is the supplied historical checkpoint, not a newly executed version 1.0.0 benchmark.

The water-limit result completes a progression from frictionless acceleration to Coulomb spreading on horizontal and inclined beds. It confirms that the underlying wet-dry and topographic terms retain their expected shallow-water behavior before avalanche-specific resistance is activated. Full numerical controls and machine-readable error summaries accompany the executable notebook, while the figure-generation notebook reads only those archived products.

B Additional WAVE verification

The two main-paper tests were selected because they most directly represent a stationary and an oscillating lake. The remaining executable notebooks cover steady hydraulic transitions, shocks, friction, a dry dam break, variable channel width, and the accelerating sloping-bed solution. Table 3 collects their principal depth errors using each benchmark’s natural one-dimensional or cross-section-mean diagnostic. The SWASHES solutions were generated from the pinned analytical implementation (Delestre et al., 2013); no numerical reference solver was used.

Figure 9 shows selected saved numerical profiles beside their analytical references. The panels cover one-dimensional hydraulic transitions, wet–dry propagation, and the full error histories for the steady Baines bump and lake-at-rest configurations.

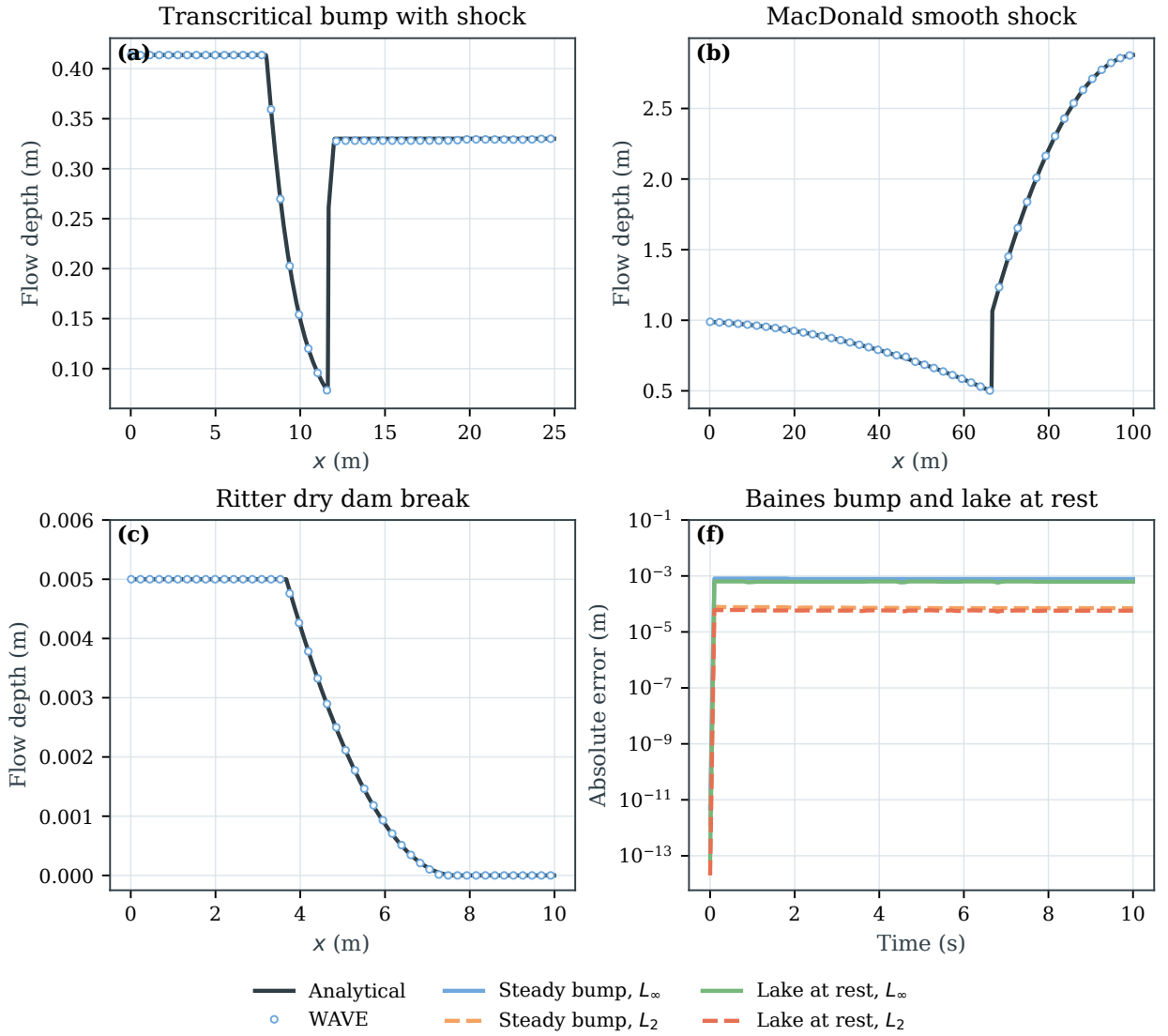


Figure 9: Additional WAVE analytical verification. (a) Transcritical flow over a bump with a shock, (b) the MacDonald smooth-transition and shock solution, (c) the Ritter dry-bed dam break, and (f) the Baines steady-bump and lake-at-rest error histories. Dark curves are analytical solutions, blue open circles are WAVE, and the pastel curves in (f) distinguish the two error norms and stationary tests.

WAVE was also applied to the frictionless dry-bottom sloping-bed solution of Ancy et al. (2008) with 0.005 m cells. Before its advancing front reached the tutorial’s open boundary, the front and rear RMSEs were 0.170 m and 0.0274 m, respectively, while the initial depth differed from the exact profile by at most 5.63×10^{-13} m. This independently exercises the water executable on the same accelerating wet–dry problem

Table 3: Additional WAVE analytical-verification results. “Centreline” applies to the two-dimensional Thacker basin, and “section mean” to the variable-width pseudo-2D channels.

Case and metric	Δx (m)	RMSE (m)	Maximum error (m)
Transcritical bump with shock, depth	0.05	4.26×10^{-3}	9.15×10^{-2}
MacDonald smooth shock, depth	0.20	7.72×10^{-3}	1.68×10^{-1}
Ritter dry dam break, depth	0.02	1.07×10^{-5}	7.15×10^{-5}
Thacker paraboloid, centreline depth	0.04	1.27×10^{-3}	2.84×10^{-3}
Pseudo-2D supercritical, section mean	0.10	3.30×10^{-2}	1.05×10^{-1}
Pseudo-2D subcritical, section mean	0.20	5.18×10^{-2}	8.36×10^{-2}
Baines steady bump, depth	0.05	7.07×10^{-5}	7.80×10^{-4}
Lake at rest, free surface	0.05	5.75×10^{-5}	6.21×10^{-4}

used for AVAC in Appendix A.

Finally, a flat-bed dry dam break compared a 0.05 m uniform grid, a 0.005 m uniform reference, and level-2 AMR with the same finest spacing. AMR reduced the RMS depth difference from 1.90×10^{-3} m to 2.30×10^{-4} m. Repeating both the fine uniform and AMR calculations with one and eight OpenMP threads produced byte-identical final native and fixed-grid files. Native non-overlapping AMR integration, rather than a coarse visualization grid, gave a maximum relative mass change of 4.68×10^{-7} across the uniform and AMR configurations.

These last two diagnostics are summarized in Fig. 10. The sloping-bed calculation tests an accelerating wet–dry boundary before the open boundary is reached, while the dam-break comparison shows the improvement obtained by concentrating the fine spacing near the front. Thread-count equality is reported separately because coincident curves alone cannot establish bitwise reproducibility.

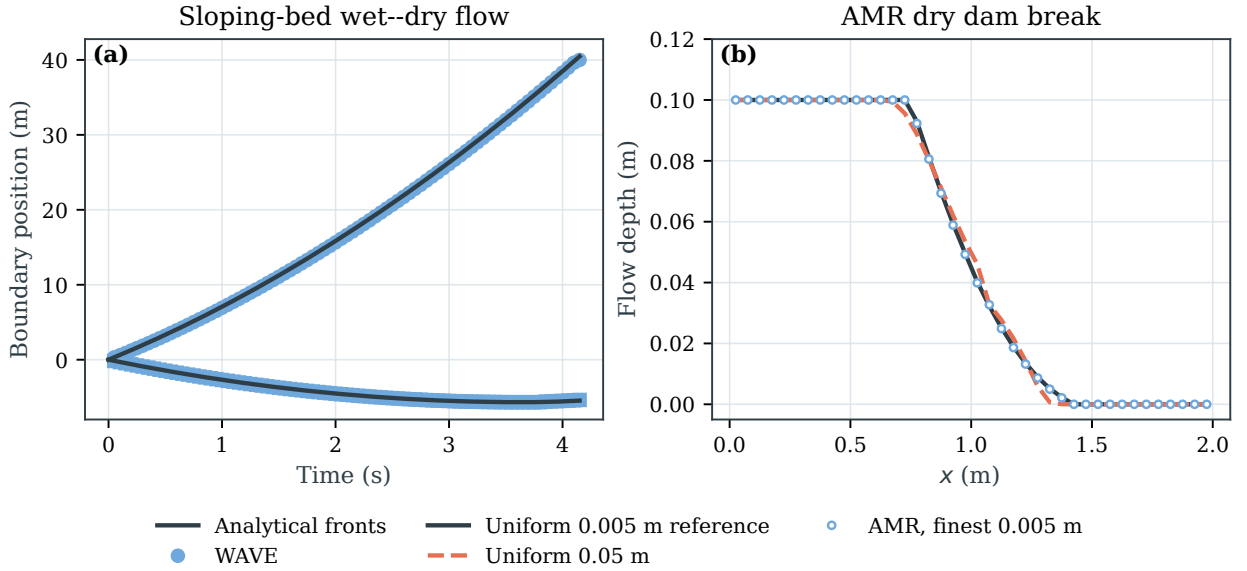


Figure 10: WAVE wet–dry and numerical diagnostics. (a) Front and rear trajectories for the frictionless sloping-bed solution before the advancing front reaches the open boundary. (b) Final dry-dam-break depth on the uniform coarse grid and level-2 AMR grid, compared with the uniform fine reference. The AMR markers use the same 0.005 m finest spacing as the reference.

C Cross-platform reproducibility

Version 1.0.0 publishes a Windows AMD64 build supporting QGIS 3.40 or newer. The preceding isolated functional workflow used QGIS 3.40.11-Bratislava: AVAC and WAVE runtimes installed automatically and an AVAC run completed. That offscreen QGIS process subsequently crashed at shutdown, so its functional pass is not a clean-exit certification. The version 1.0.0 packaging and isolated-runtime installation checks are separate from that earlier GUI run. No new complete scientific suite or interactive QGIS session was run solely to change the version label.

The AVAC and WAVE numerical executable SHA-256 values are, respectively,

```
55a8c4baf0cc74d9a8ca4c582f361eb85ab01817b0d16b63e1c824611144f2e3  
05bbcc28e71554cf13d1cc76c05329af1eb42a2966a8ec4738ef0310b9cbc555.
```

They are unchanged from the accepted Windows 0.6.1 package; runtime-version metadata and plugin release descriptors are updated to 1.0.0. Clawpack is 5.14.0 and the required Windows libraries and component licences are bundled. The package checksum and full runtime inventories accompany the release. Previous macOS builds remain historical releases, and no macOS or Linux 1.0.0 binary or new cross-platform scientific-equivalence claim is supplied by this Windows promotion.

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